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Transformer Protection – Incidents and Lesson Learnt

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Transformer Protection – Incidents and Lesson Learnt

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132/11kV Transformer Protection

Main Protection

- 2 x Biased differential protection (BD)
- 1x HV restricted earth-fault protection (HVREF)
- 1 x LV restricted earth-fault protection (LVREF)
- High voltage winding instantaneous high set over-current (HSOC)
- Buchholz gas surge protection (Buchholz)
- Winding and oil temperature protection (WT/OT)

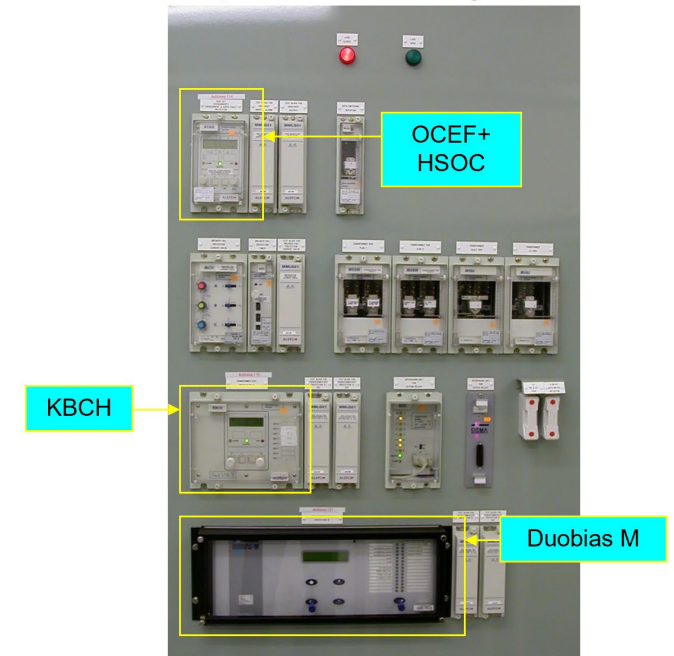
Backup protection

- Over current & earth fault protection (OCEF) at 132kV source
- For 35MVA Tx in parallel
 - Directional OCEF protection (DOCEF)

Traditional transformer protection relays

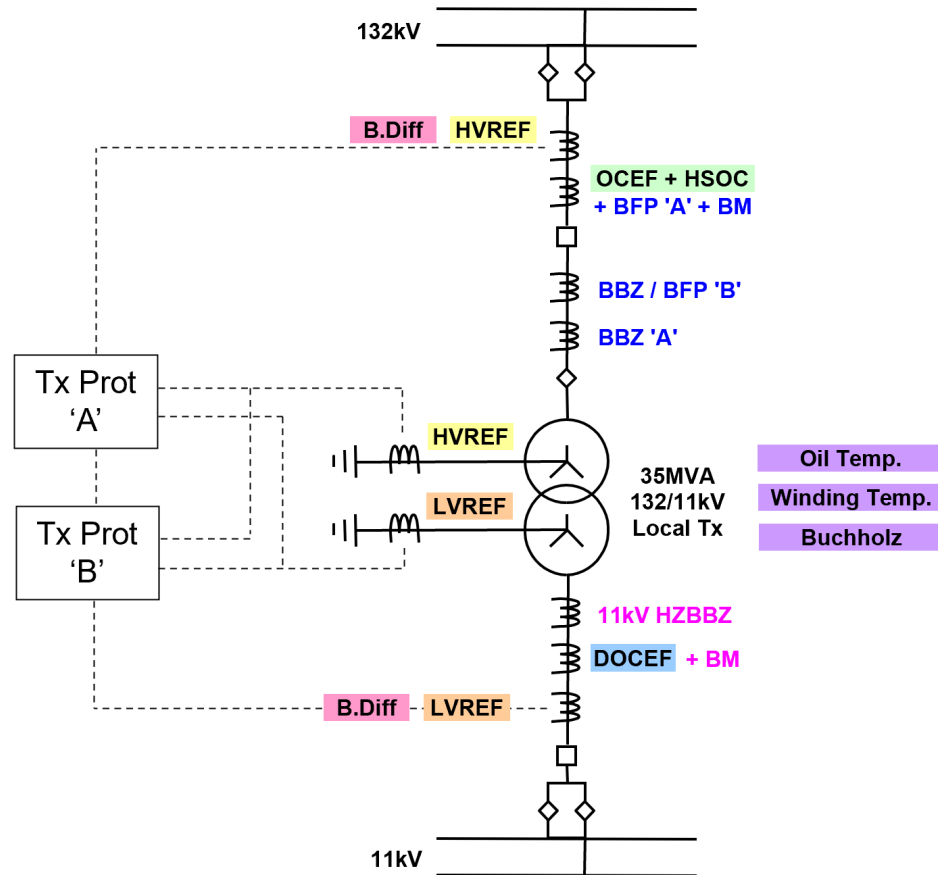


Integrated digital protection relays

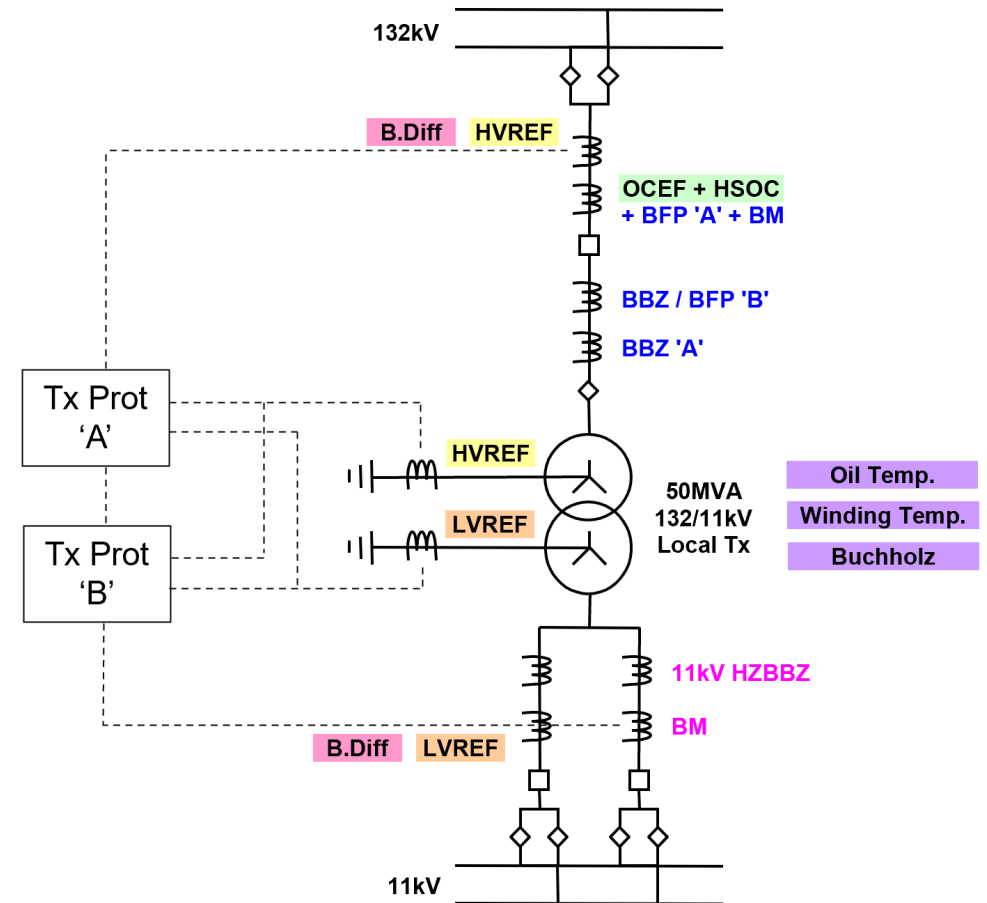


132/11kV Transformer Protection – 35MVA vs 50MVA

35MVA Tx -



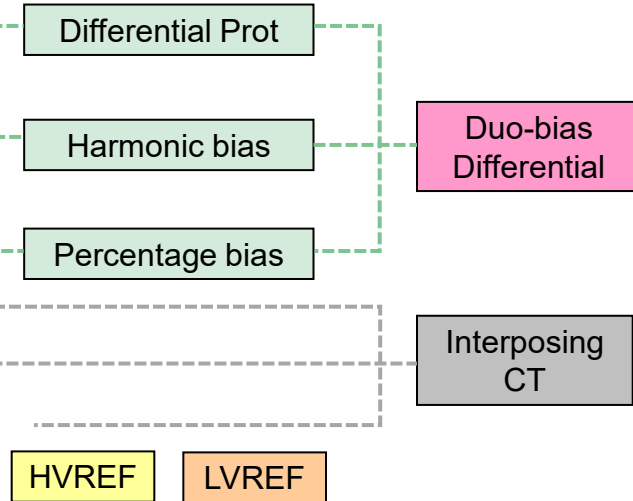
50MVA Tx -



132/11kV Transformer Protection – Design

Main protection

- Unit protection
- Able to cater for the following situations
 - Magnetizing inrush current
 - Current mismatch
 - Tap changer
 - CT ratio
 - Transformer ratio & phase shift
 - Flow of zero phase sequence current
 - Inadequate earth fault sensitive

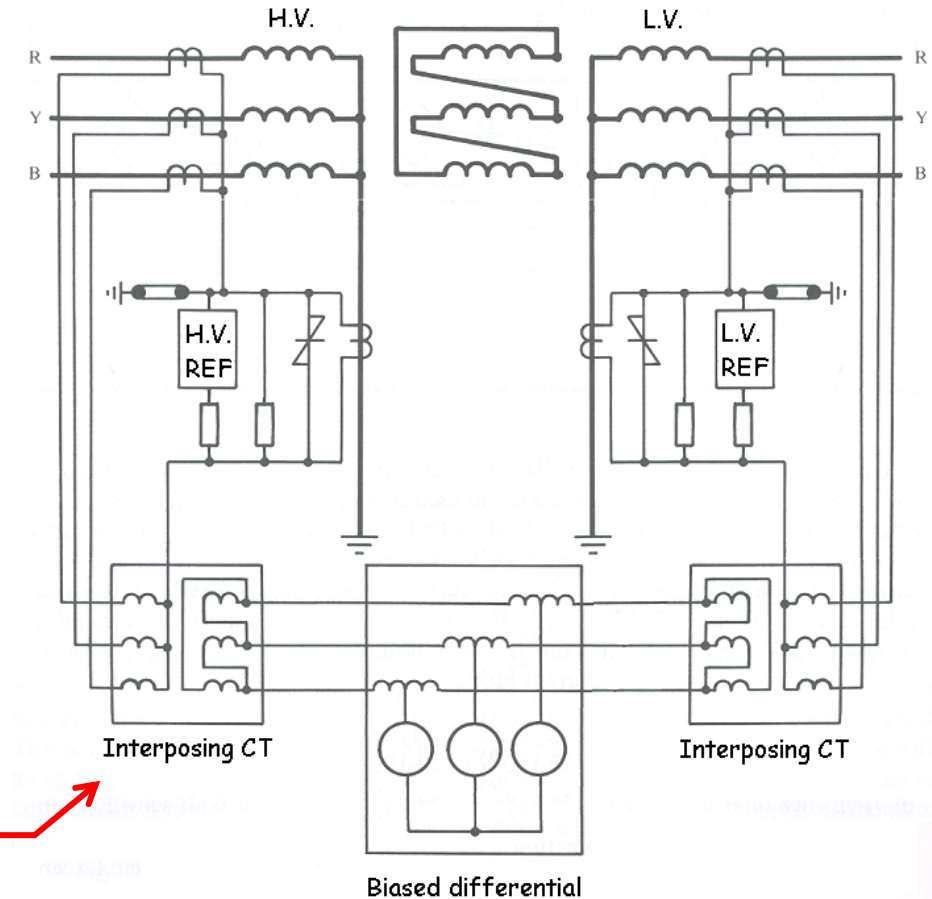


Backup protection

- OCEF + DOCEF

Additional protection

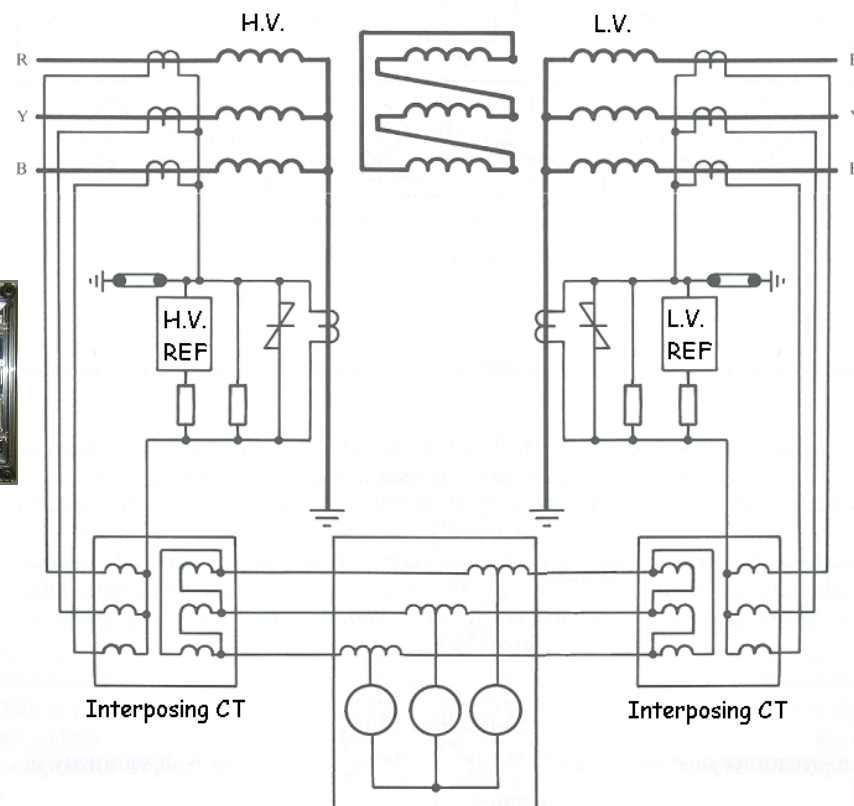
- Oil Temp / Wind Temp Trip + Buchholz Surge Trip (Tx / Tc)



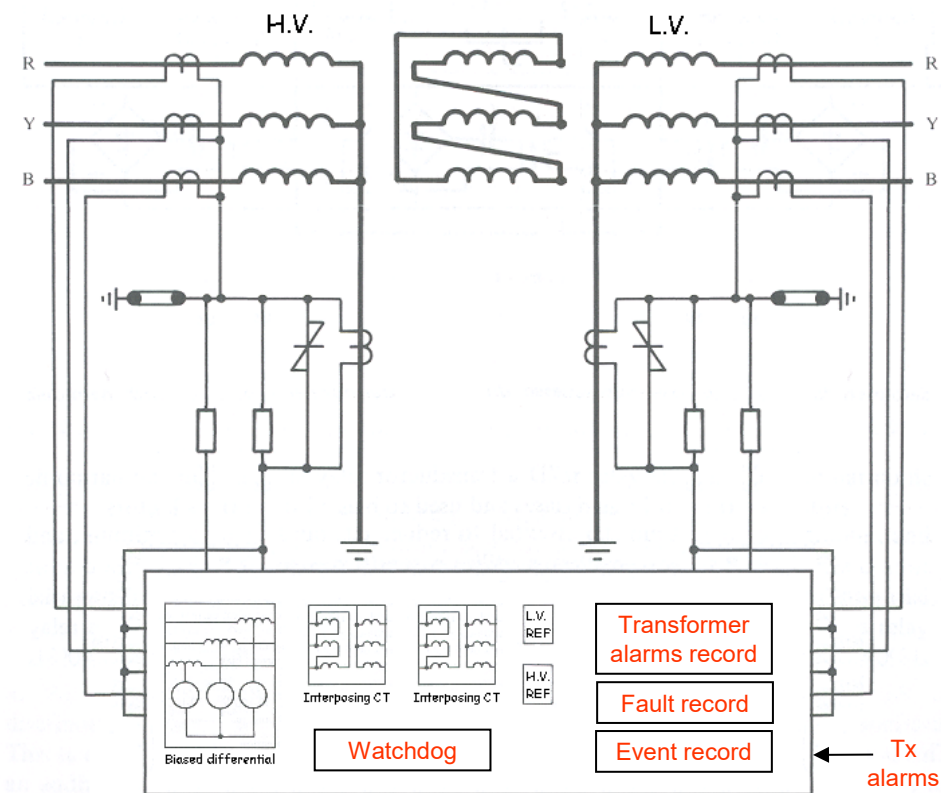
- For ratio correction between HV & LV
- For the formation of delta circulating loop for differential purpose

132/11kV Transformer Protection – From EM to Digital

FAC14 / 4B3



4C21



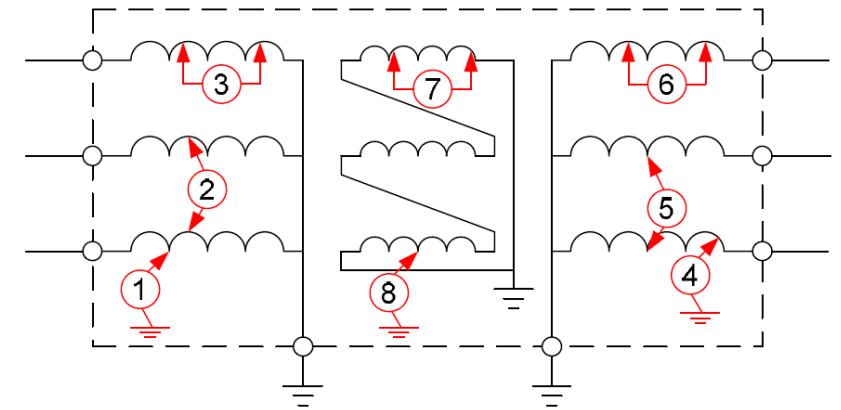
KBCH



Duobias - M

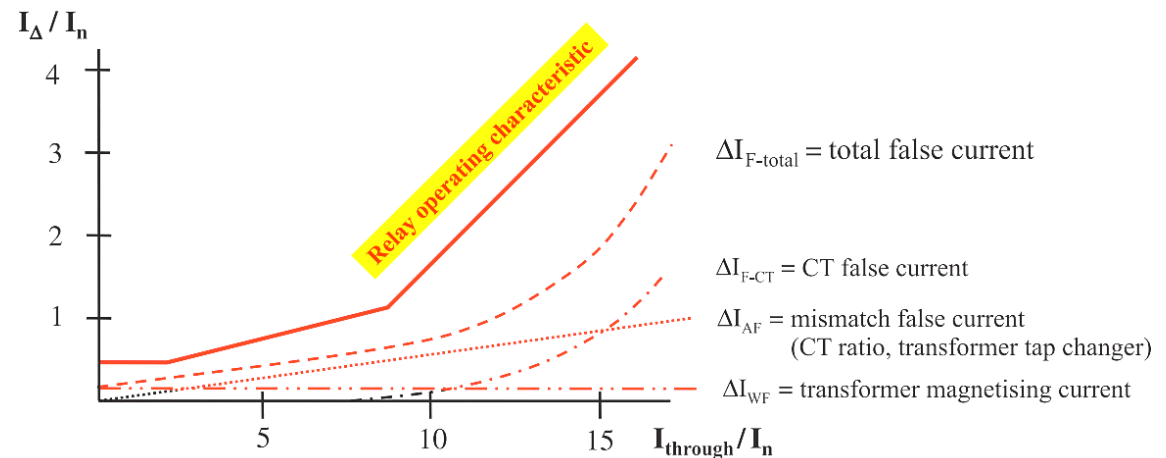
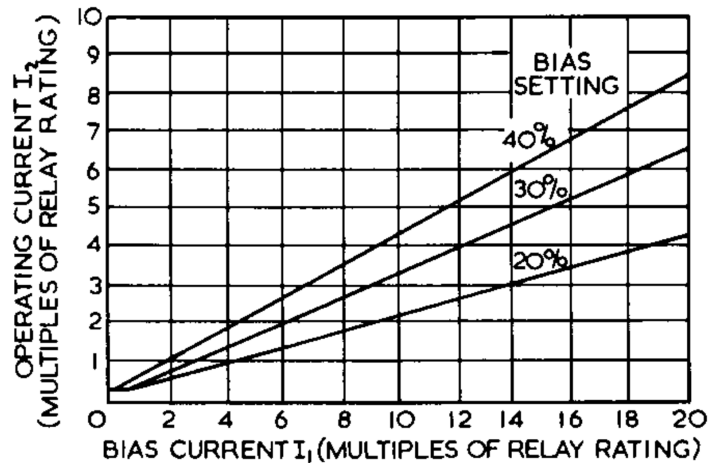
132/11kV Transformer Protection – Protection Summary

Fault Type	Protection Used
Pri. winding ph-ph fault	Differential, OC+HSOC
Pri. winding ph-E fault	Differential, HVREF, EF
Sec. winding ph-ph fault	Differential
Sec. winding ph-E fault	Differential, LVREF
Interturn/Core/Tank fault	Buchholz
Overheating	Oil & winding temp.
Reverse current	DOCEF (35MVA Only)



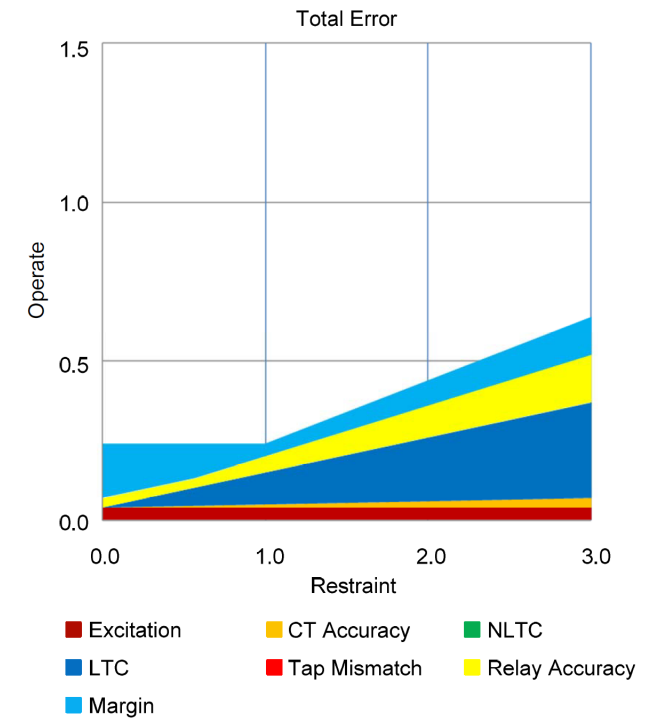
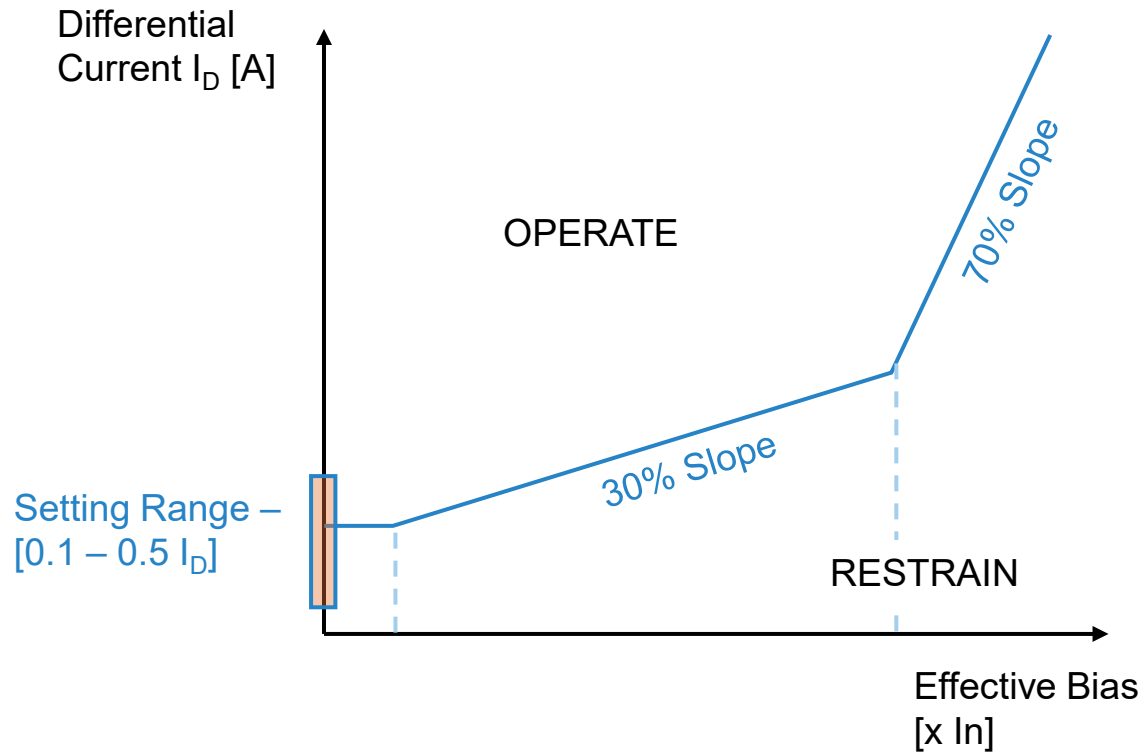
132/11kV Transformer Protection – Percentage Bias

- (1) **Percentage Bias** – maintain **stable for full load max/min tap with CT error and through fault condition** while remaining **sensitive for internal fault** with percentage differential plane (in multiple sections).
- **Steady State Differential** = Steady State Magnetizing Current
 - **Percentage (a.k.a. Proportional) Differential** = Tap Mismatch + CT Error (under nominal load / through fault)
 - **Transient Differential** = CT Saturation (due to high secondary voltage = symmetric saturation; due to DC offset = asymmetric)
 - Line CT/ICT ratios and correction factors set to achieve current balance (1A into relay) at **nominal tap**
 - Off nominal tap creates a differential, but it should be well below operating characteristics.
 - Mal-operation due to this cause is avoided by selecting the **minimum bias** to be greater than sum of the maximum tap of the transformer and possible CT errors



132/11kV Transformer Protection – Percentage Bias

Percentage Bias Curve -



Bias in P643 to provide higher through fault stability:

- **Delayed Bias:** maximum bias quantity within last cycle.
- **Transient Bias:** additional bias introduced with sudden increase in mean-bias, decayed exponentially.
- **Maximum Bias:** maximum bias quantity in three phase[#]

First section is set higher than transformer magnetising current

(at steady state)

Second section is set to allow for off-nominal tap settings

Third section has a larger bias slope to cater for heavy through-fault conditions

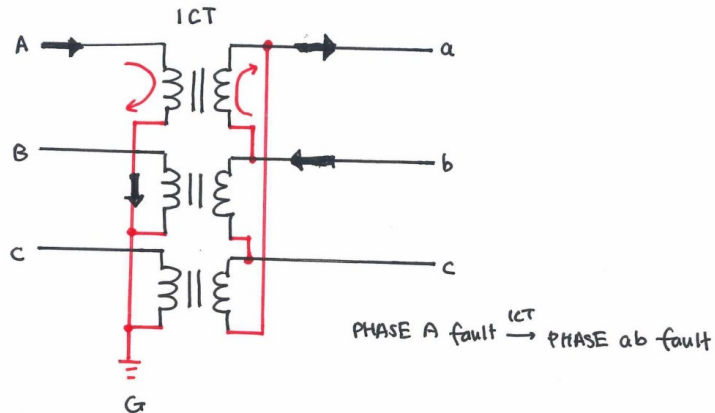
[#] MAX $\{I_a, I_b, I_c\}$ neglects any CT saturation in phase.

132/11kV Transformer Protection – Interposing CT (ICT)

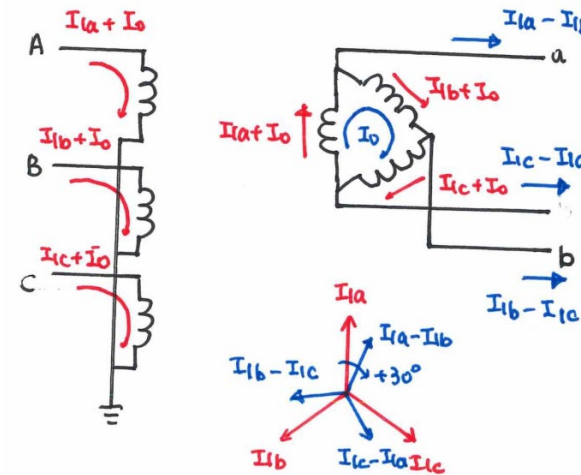
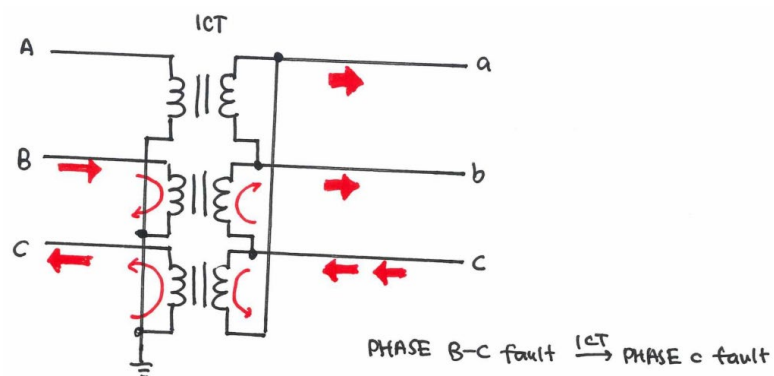
Interposing CT is intended to correct CT ratio, balance phase shift and filter out zero-sequence current.

In the case of a Y-delta transformer, an external fault will cause zero sequence current to be measured by the Y-side CTs. However, the zero-sequence current is trapped on the delta side, it is not measured by the delta-side CTs.

Earth Fault:



Phase Fault:



Zero Sequence Filtering

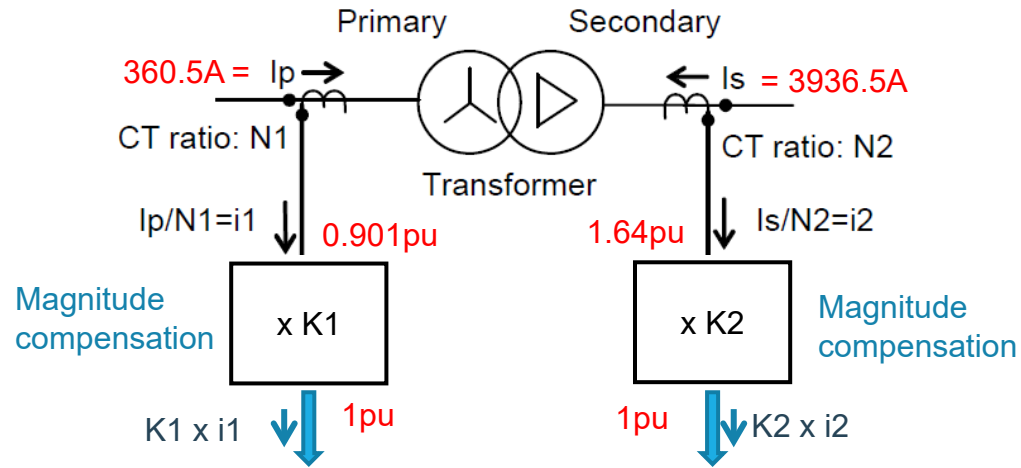
From WYE to DELTA with 30° clockwise:

$$\begin{pmatrix} I_a' \\ I_b' \\ I_c' \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} I_a \\ I_b \\ I_c \end{pmatrix}$$

Zero Sequence Filtering:

$$\begin{pmatrix} I_a' \\ I_b' \\ I_c' \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \begin{pmatrix} I_A \\ I_B \\ I_C \end{pmatrix}$$

132/11kV Transformer Protection – ICT Corr Factor in Digital Relay



For 132/11kV 75MVA Tx with $Z_{TX} = 42.58\%$, tap range +10.5 ~ - 28.5%

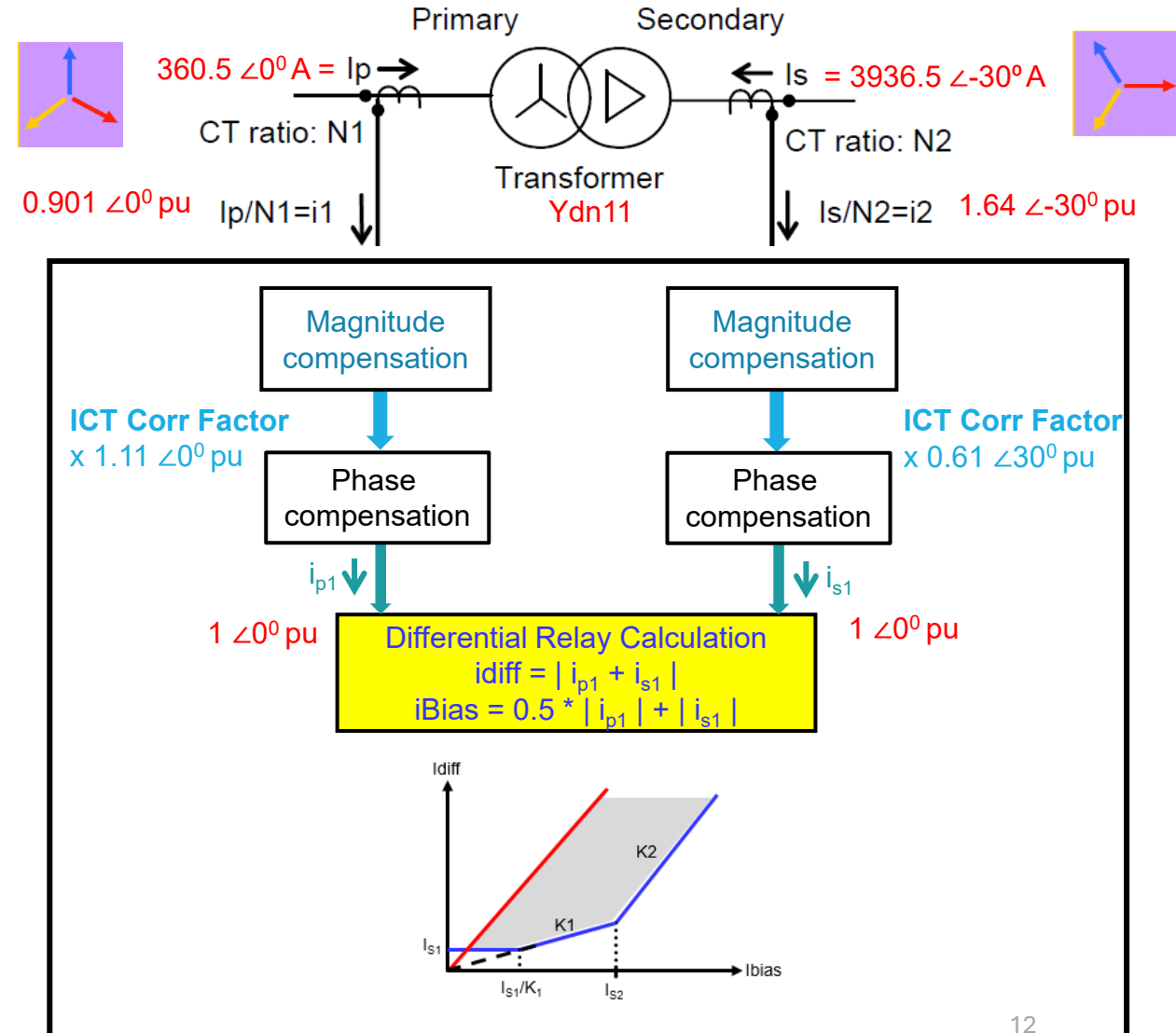
For **Nominal Load Condition** @Mid tap:

$$I_{FL_HV} = \frac{75MVA}{\sqrt{3} \times 132kV(1 + \frac{10.5\% - 28.5\%}{2})} = 360.5A_{PRI} = 0.901A_{SEC}$$

Magnitude Compensation Factor $K1 = 1A / 0.901A = 1.11$

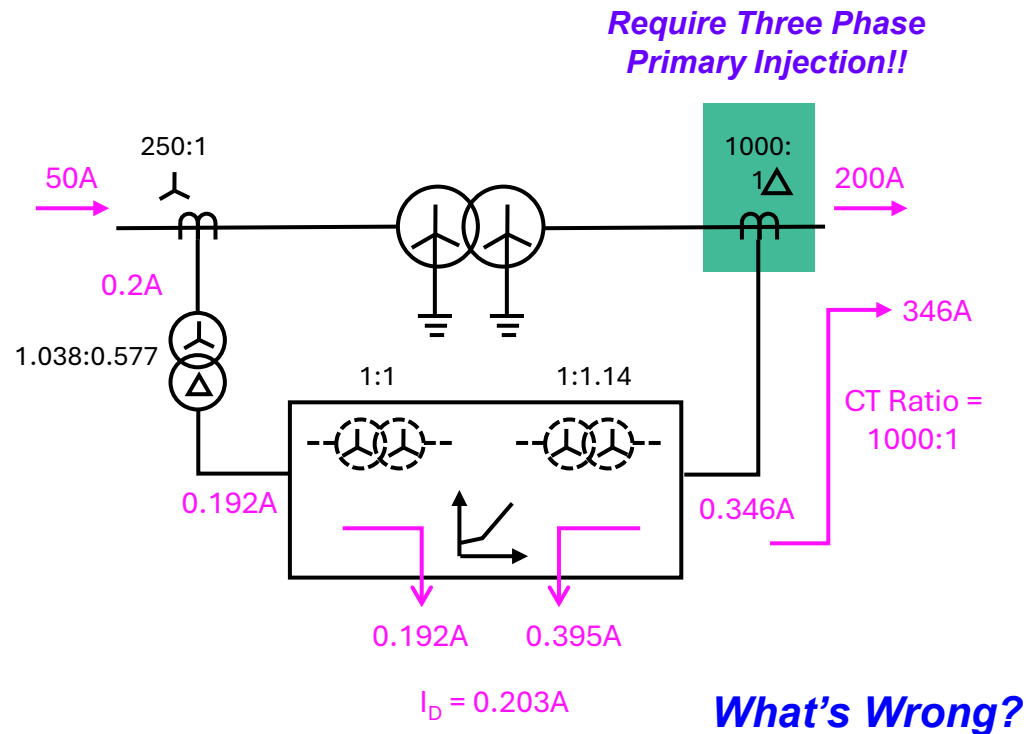
$$I_{FL_LV} = \frac{75MVA}{\sqrt{3} \times 11kV} = 3936A_{PRI} = 1.64A_{SEC}$$

Magnitude Compensation Factor $K2 = 1A / 1.64A = 0.61$

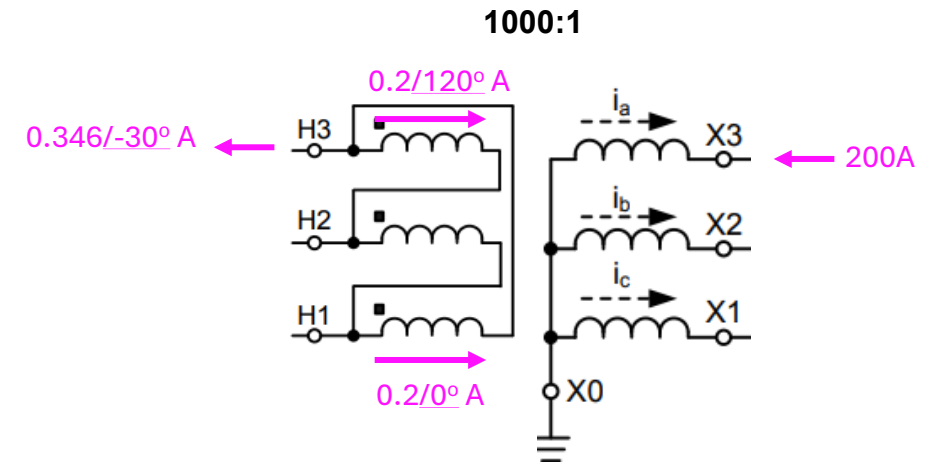
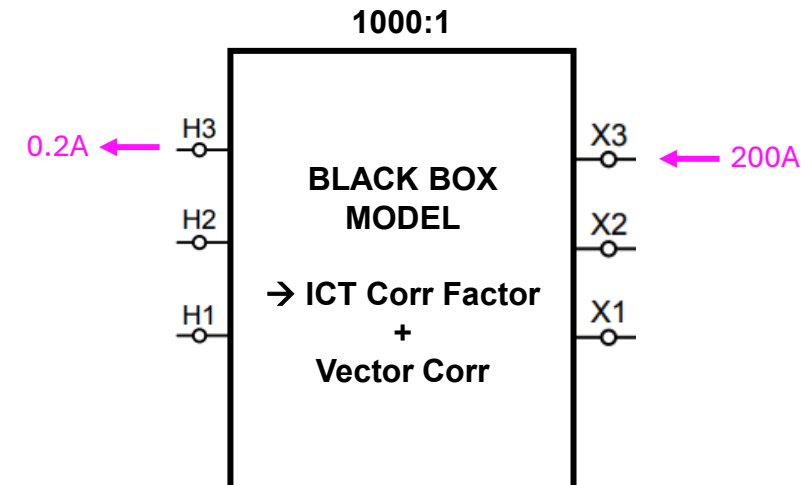


Case 1: Incorrect Setting Leading to Tripping

- Prior to the incident, the circuit were restored to service at 04:35 after 33kV cable circuit was diverted to a temporary switchboard. It was found with differential current under on load test, but the test result was confirmed with satisfactory. At 06:50, the circuit was tripped with incorrect setting.



CT Manufacturer:

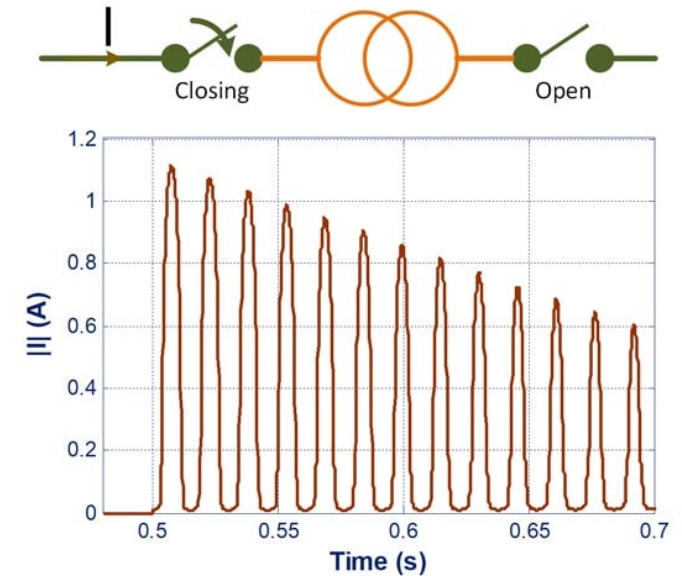
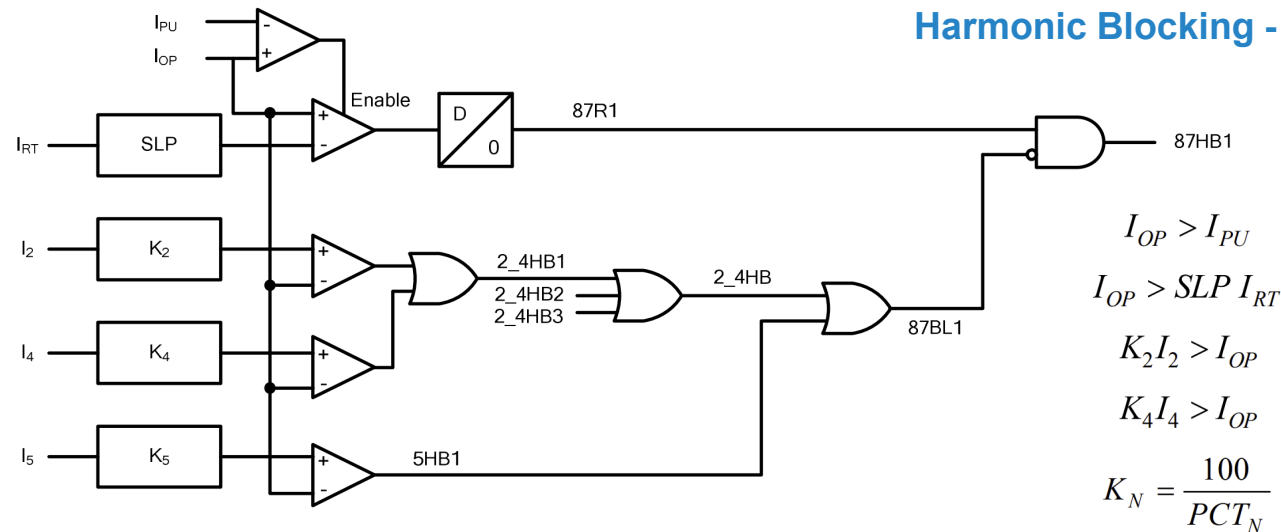
**Relay Manufacturer:**

132/11kV Transformer Protection – Harmonic Bias

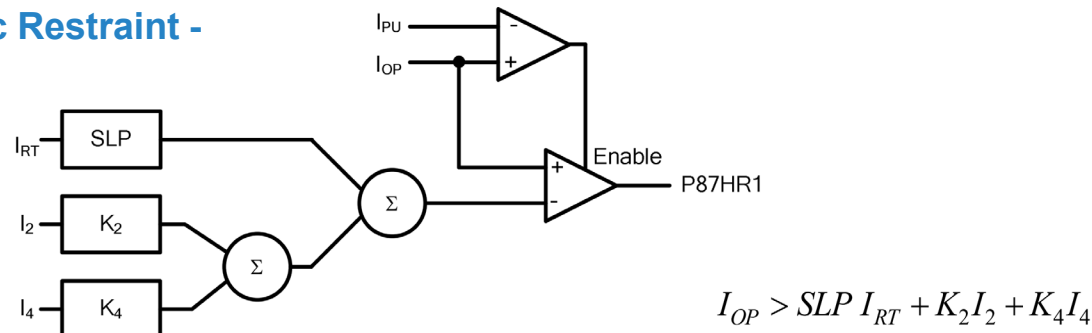
(2) **Harmonic Bias** – prevent relay operation due to flow of pulses of **magnetizing inrush current** into one winding when the transformer is first energized

Note – 1) It is important to **detect fault upon energization**, so that the harmonic restraint function or blocking function should not stop fault detection and clearance.

2) 4C21 employs **cross-phase harmonic restraint**; while KBCH & DBM employs **harmonic blocking**.



Harmonic Restraint -



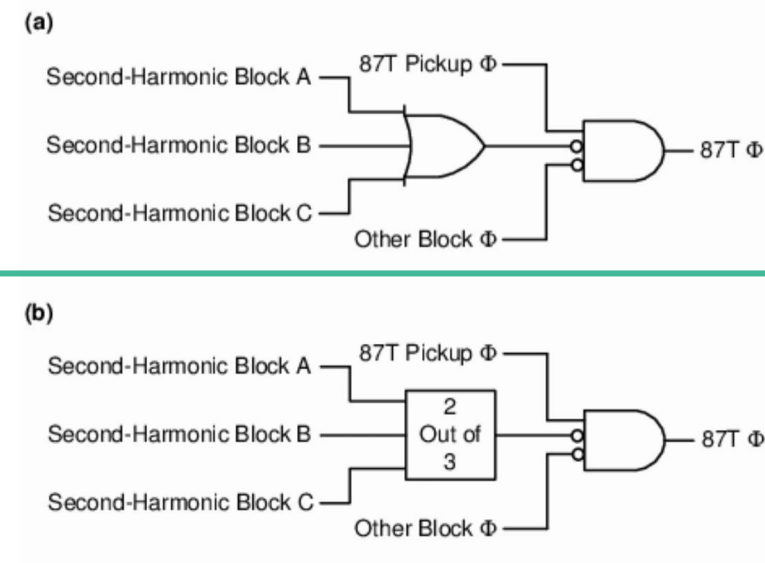
Self-Adaptive Blocking vs Phase-to-Phase Cross Blocking

Self-Adaptive Blocking

- Automatically decreases the restraint coefficient of harmonics during the initial stage of transformer energization, and with the passage of time automatically increases those values which shall be not greater than corresponding values of settings [87T.K_Hm2_Inrush]. This feature ensures biased differential protection fast operation for transformer energized on to a fault.
- If inrush current is distinguished, equipment enables phase-to-phase crossing block mode for short time, and then switches to phase-segregated blocking mode after a period.
- If the second harmonic content continues to rise, then biased differential protection is kept being blocked.

Phase-to-phase Crossing Blocking Mode

- Two-out-of-Three Logic:** If two phase current are distinguished not inrush current, the differential element of the two phases will be released to operate.

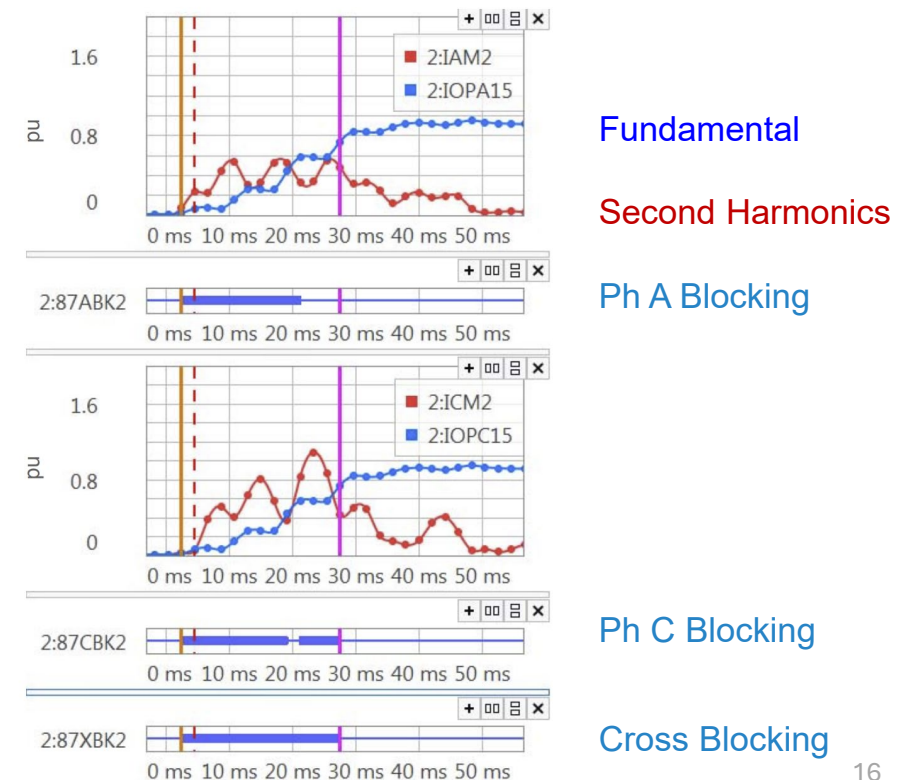
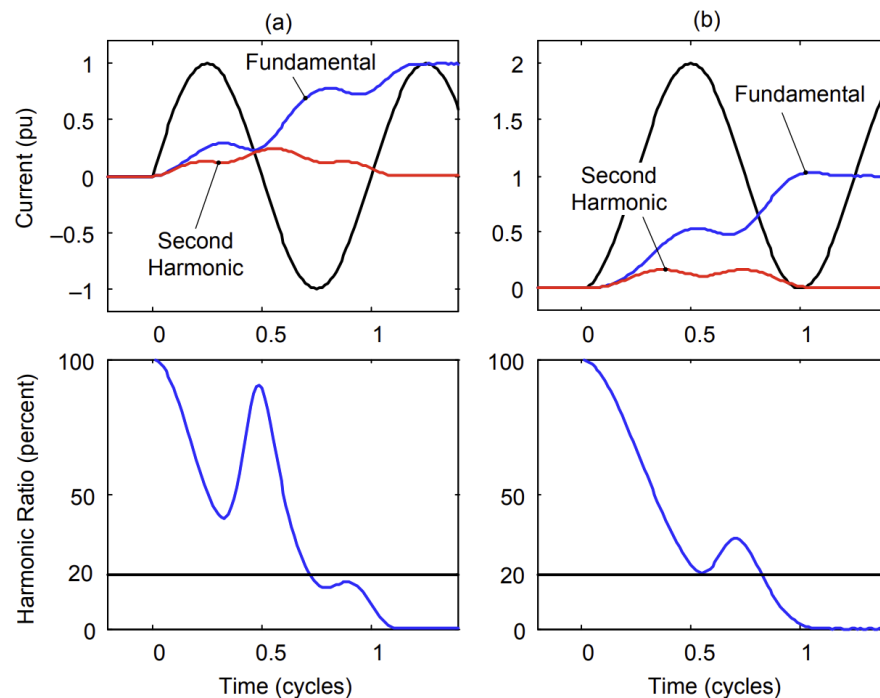


Situation	Harmonic of each phase current	Mode 1: self-adaptive blocking mode	Mode 2: phase-to-phase crossing block mode
1	I _{da} (great harmonic)	Diff prot not operate	Diff prot not operate
	I _{db} (great harmonic)		
	I _{dc} (great harmonic)		
2	I _{da} (little harmonic)	Diff prot operate	Diff prot operate
	I _{db} (little harmonic)		
	I _{dc} (little harmonic)		
3	I _{da} (great harmonic)	Diff prot may operate or not according to differential current waveform	Diff prot not operate
	I _{db} (great harmonic)		
	I _{dc} (little harmonic)		
4	I _{da} (great harmonic)	Diff prot operate	Diff prot operate
	I _{db} (little harmonic)		
	I _{dc} (little harmonic)		

132/11kV Transformer Protection – Harmonic Bias

Problem of Harmonic Blocking – Spurious Transient Output with Second Harmonic Filter leading to Blocking on Fault

- The **rise of differential current** at the inception of an internal fault generates **spurious transient output** from **harmonic filters**, which can slow down the 87R element.
- The second-harmonic filter is excited by the changing input signal, taking approximately **one cycle to settle** to a true value of zero. During this period, the **second-harmonic ratio** is large, potentially **blocking** or **restraining** the differential element.
- The figure shows the delay in operation of the 87R element during a **heavy internal fault** due to harmonic methods used for security on inrush. The second-harmonic magnitudes (red traces) and fundamental magnitudes (blue traces) illustrate the blocking condition.

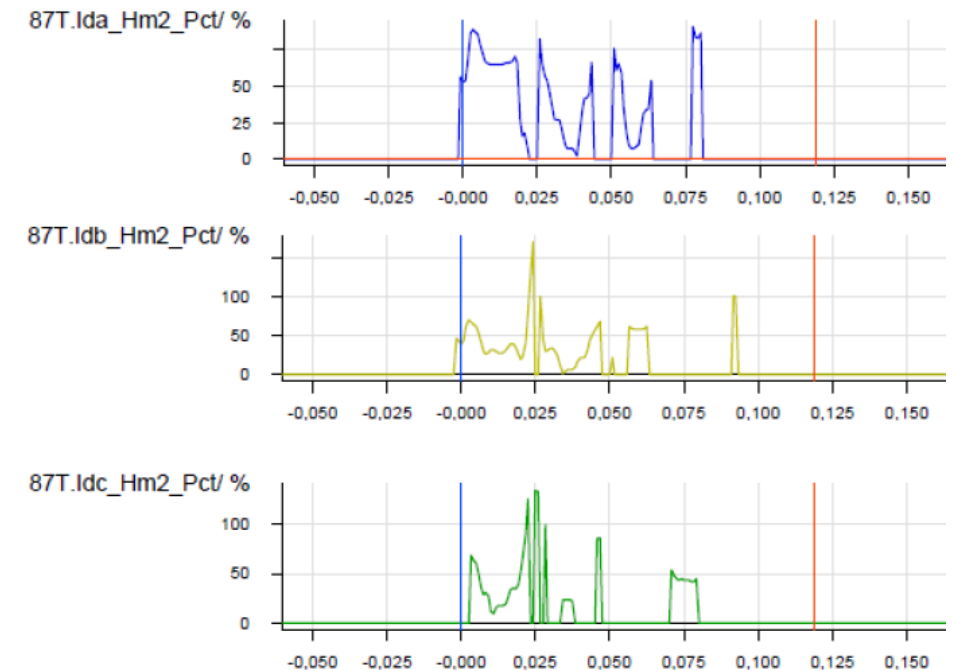
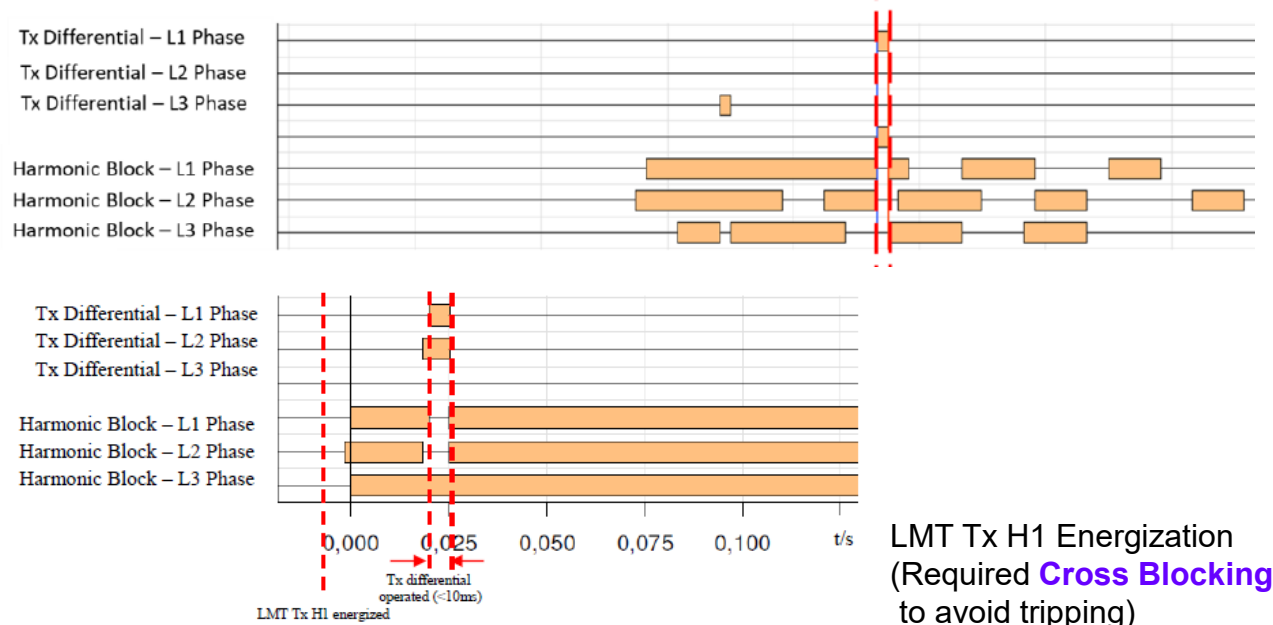


Case 2 – Transformer Tripped under Energization and De-Energization with Harmonic Blocking

- Upon de-energization of a transformer, Tx differential relay 'A' operated, and Intertripping was initiated.
- Both transformer differential relays detected oscillation during transformer de-energization. second harmonic content of the oscillation was marginal relative to the harmonic restraint setting in use (<20%), and there was a short duration (<5 ms) where the **harmonic blocking elements** reset for all 3 phases
- Similar Case under Energization occurred on 2018-06-07.

Follow Up Action

- Set harmonic restraint setting of both transformer differential relays to **15%**, especially for circuits with relatively long cables and circuits whose pre-commissioning tests were conducted before May 2019.



External Fault Detection with Transient Biasing

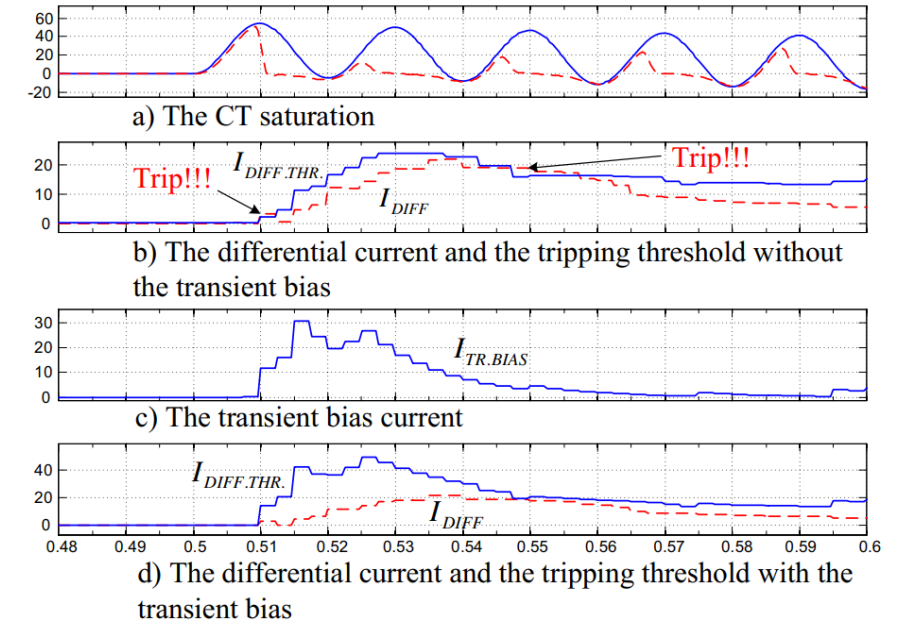
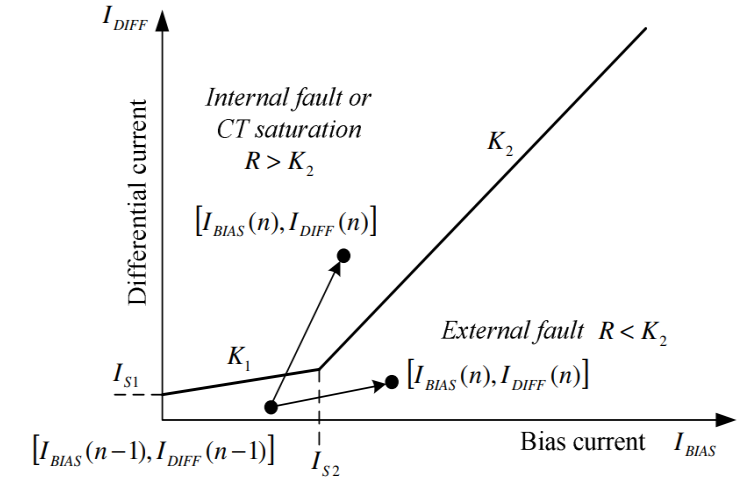
Transient bias is added to prevent false tripping of the protection system during external faults with CT saturation. When a **transient condition** is detected, the transient bias algorithm temporarily increases the tripping threshold of the differential protection.

1. This increase is based on the **positive ΔI_{BIAS}** . If the bias current increases, the tripping threshold is raised to prevent false trips.
2. The transient bias **decays quickly** once the transient condition subsides, ensuring that the protection system returns to normal sensitivity for detecting internal faults.

$$\begin{aligned}
 &\text{If } R < K_i \text{ and } \Delta I_{BIAS}(n) > 0 \\
 &\Delta I_{DIFF}(n) = I_{DIFF}(n) - I_{DIFF}(n-1), \\
 &\Delta I_{BIAS}(n) = I_{BIAS}(n) - I_{BIAS}(n-1), \\
 &R = \Delta I_{DIFF}(n) / \Delta I_{BIAS}(n). \\
 &\quad \quad \quad I_{TR.BIAS}(n) = D \cdot I_{TR.BIAS}(n-1) + S \cdot \Delta I_{BIAS}(n), \\
 &\quad \quad \quad \text{otherwise} \\
 &\quad \quad \quad I_{TR.BIAS}(n) = D \cdot I_{TR.BIAS}(n-1),
 \end{aligned}$$

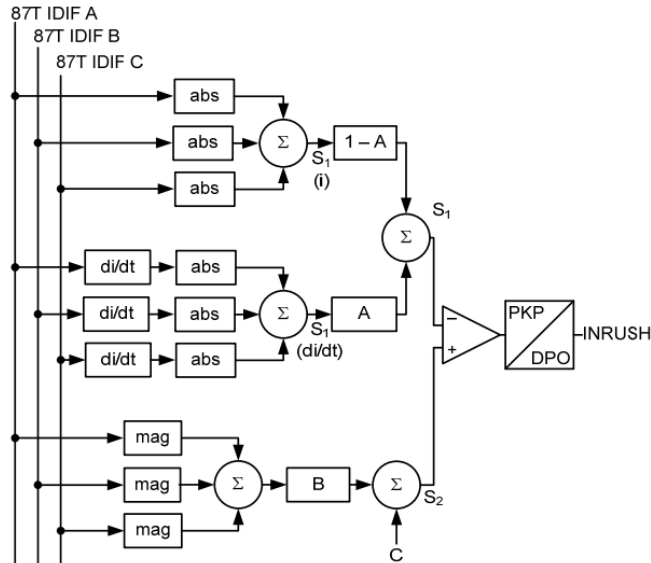
External Fault Detector (EFD) aims to prevent false trips during external fault and ensure quick and accurate detection of internal faults by maintaining the sensitivity of the differential protection.

- The system uses the ratio $R = \Delta I_{DIFF} / \Delta I_{BIAS}$ to detect external faults. If the ratio indicates an external fault, the transient bias is applied to stabilize the system.



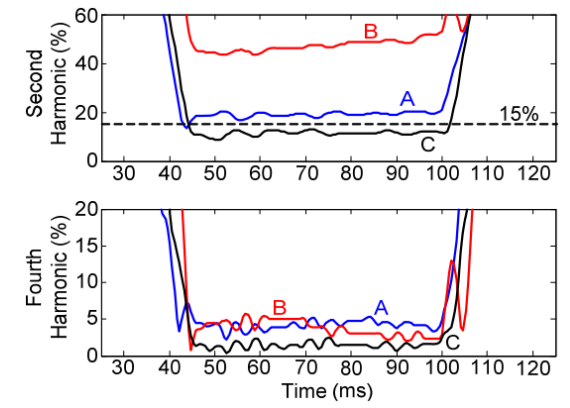
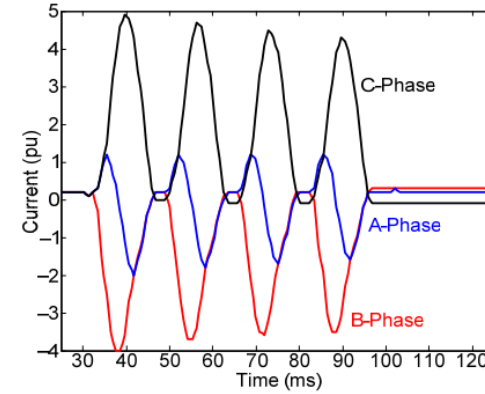
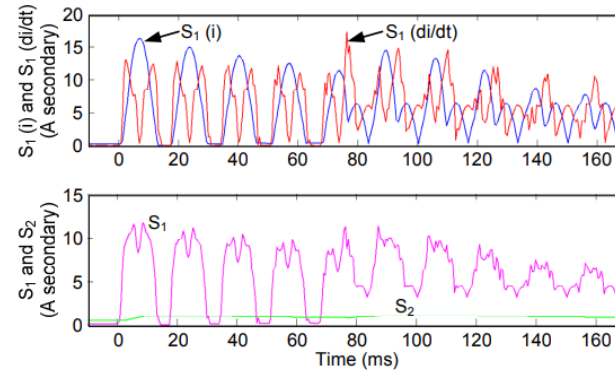
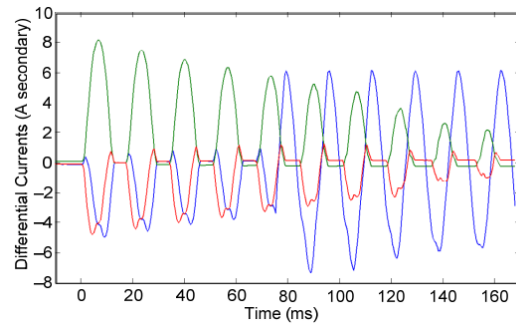
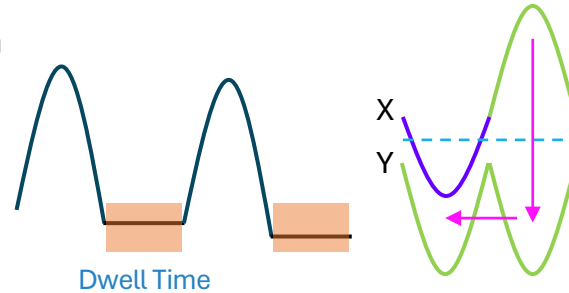
Inrush Detection

Dwell Time based Inrush Detection -

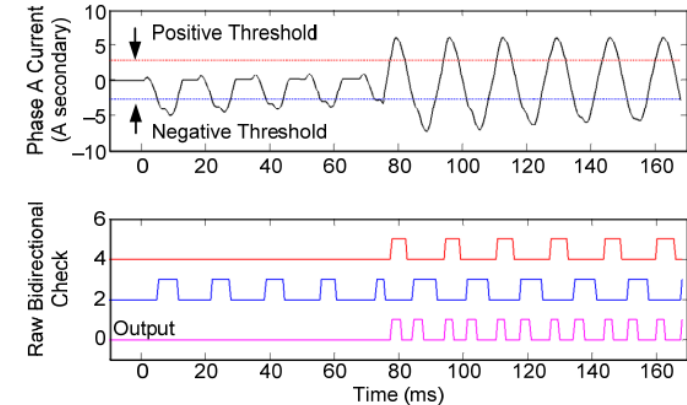
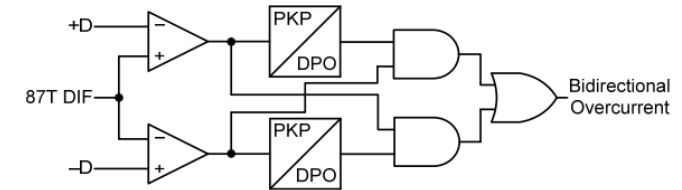


No Gap Detection: counts the numbers of consecutive samples below a dynamic threshold to determine the low current gap duration in the waveform (P643)

Wave Distortion Detection: determine the asymmetry with $K = |S_+| / |S_-|$, where $|S_+|$ is area of $X - Y$ and $|S_-|$ is area of $X + Y$. (PCS978)

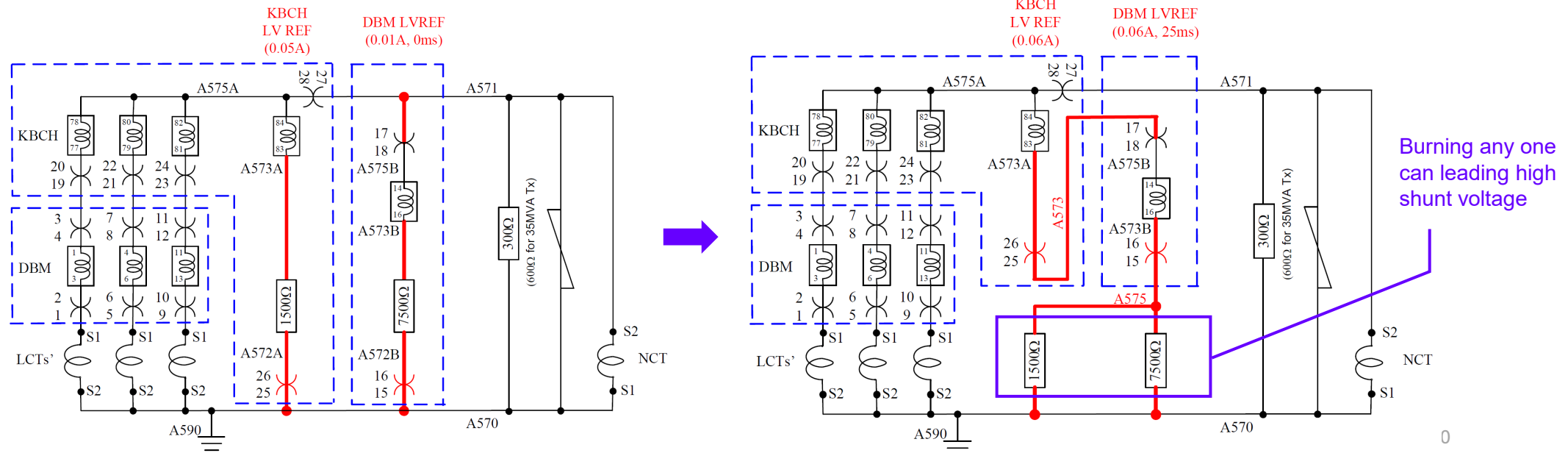
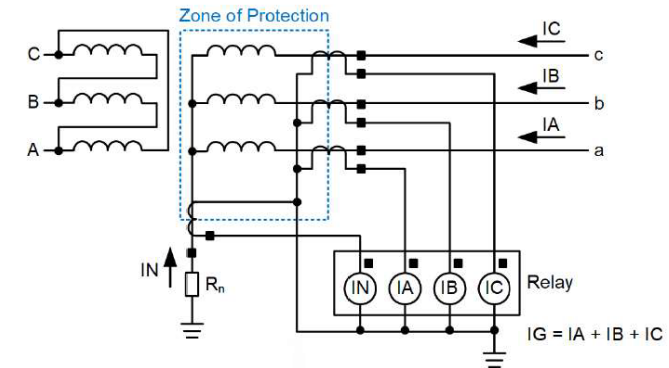


Bidirectional Overcurrent based Inrush Detection -



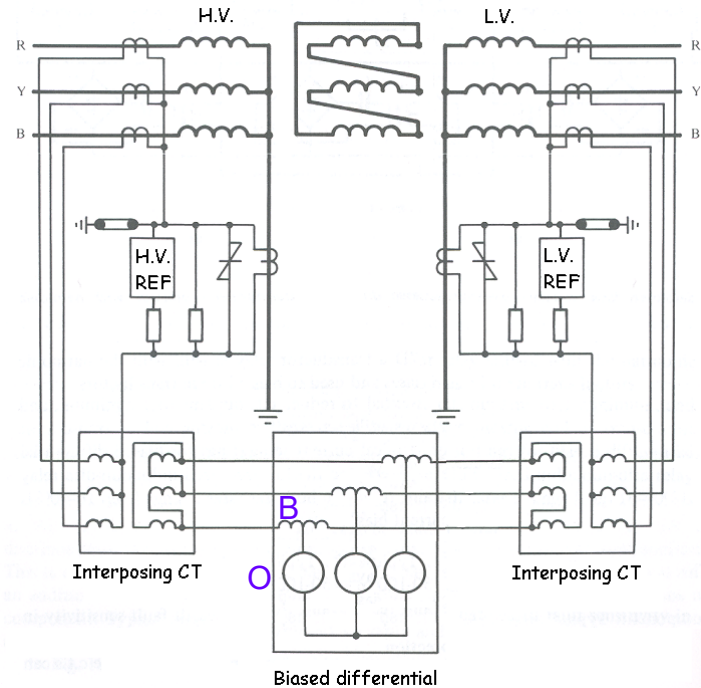
132/11kV Transformer Protection – REF in High Impedance Principle

- After replacing the standard high impedance EM relay (4C21 for duo bias + FAC for REFs), the scheme with digital relays (KBCH + DBM; not applicable to GRT100 + PCS978) is in **high impedance principle** as required in PSS 3-1: 2008.
- Restricted Earth Fault (REF)** Protection is to enclose a “**restricted area**” with CTs such that the current sum should not be large under normal condition. However, with the **mismatch in line CT and neutral CT**, it is often that the neutral CT become saturated, hence leading to a differential current, and it is solved by high impedance principles.
- As illustrated below, another consideration is whether to put the REF element “IN” in **series** or in **parallel** after summing up the current. It was noted that even if the resistance was designed to obtain the **current ratio** to trigger the REF element of KBCH and DBM accordingly, the current flows between two parallel paths becomes not as expected with **resistor degradation**. Hence, the scheme was made series after that.



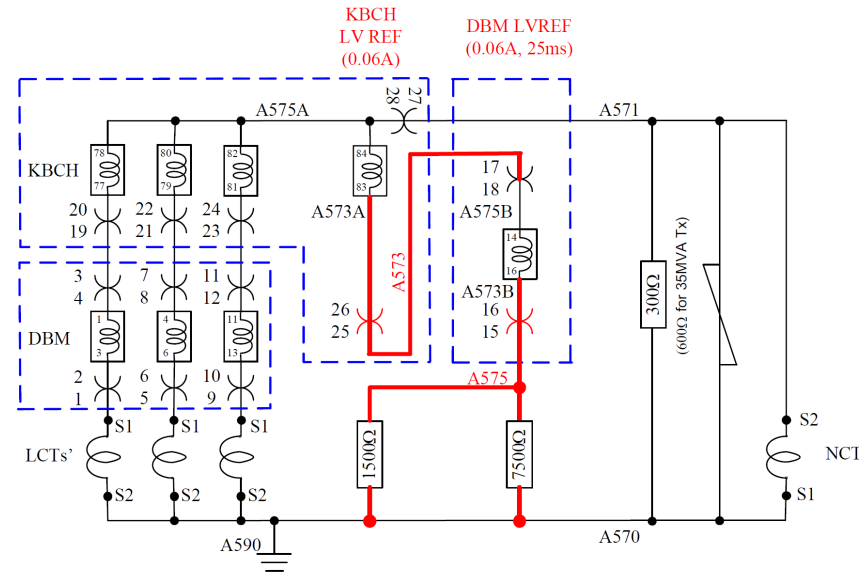
Coordination between DB Element and REF Element

4C21 + FAC14



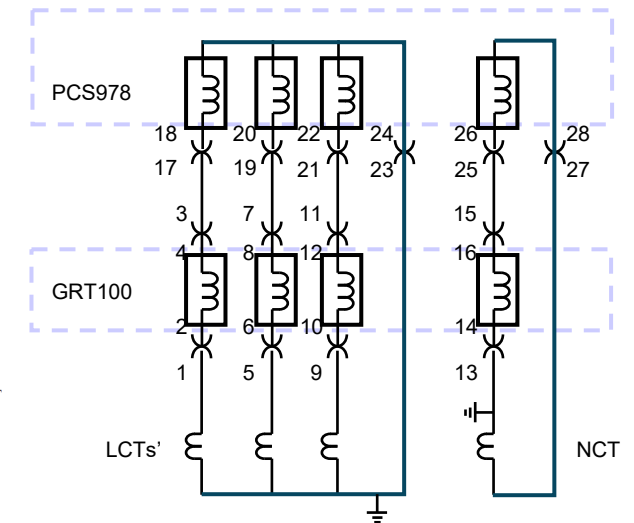
REF with **high impedance scheme** to sum up current with parallel CT. Phase current is that replicated with ICT (and filter zero seq with delta connection) to go through bias and op coil for DB.

KBCH + DBM



Phase CTs go through KBCH and DBM for DB calculation first, and to be summed up with parallel CT in **high impedance principle** and go back to the REF element in KBCH and DBM in series, which acts as an OC element ($I > I_{SET}$).

GRT100 (P643) + PCS978



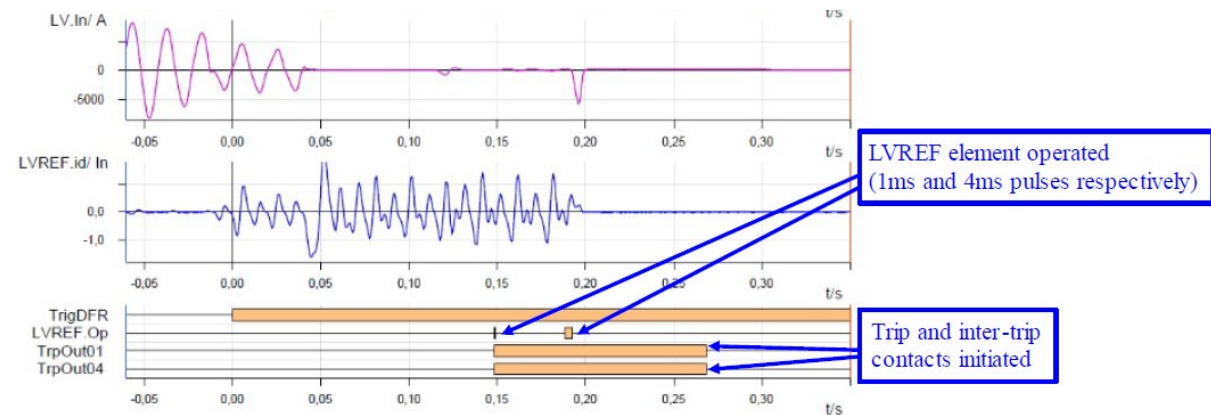
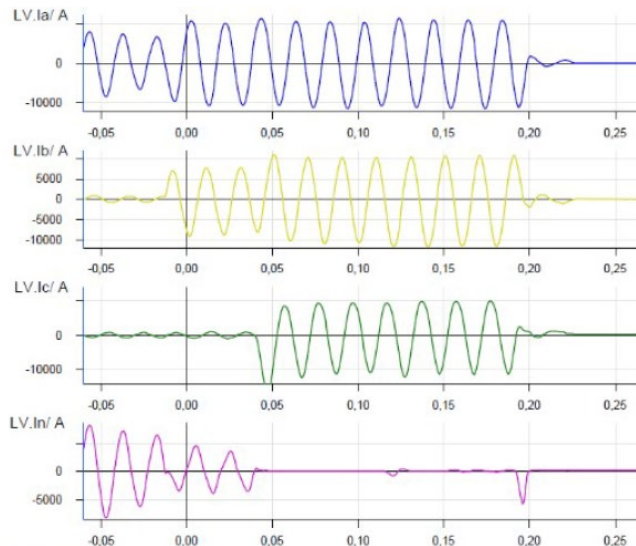
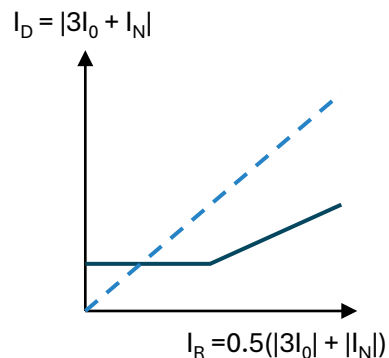
Phase CTs and N CT goes into GRT and PCS relay separately in **low impedance principle**. DB and REF are calculated with sampled data.

Case 3: LVREF Unstable Leading to Through Fault Tripping

- On 19-Sep-2023 at 15:34hr, a transformer tripped simultaneously under **through fault** with **LVREF element**.
- From the fault records in one relay, current measurement (both 3-phase and neutral) of the relay were normal and found matched with those from the other relay and fault recorder. However, it was observed that the **LVREF element (low impedance, percentage differential type)** of one transformer differential relay operated twice at very **short pulses** of 1ms and 4ms respectively.
- It is concluded that the mal-operation of the relay should be caused by the relay internal logic on LVREF detection upon high level of **through fault current**.

Follow Up Action

- Change the **delay timer setting** for LVREF to 25ms to ride-through any inrush.
- Apply blocking function (“**Harmonic Blocking of REF**” and “**Positive Sequence Current Restraint Block**”) through Firmware Upgrade from V1.11 to V1.12 for new projects.

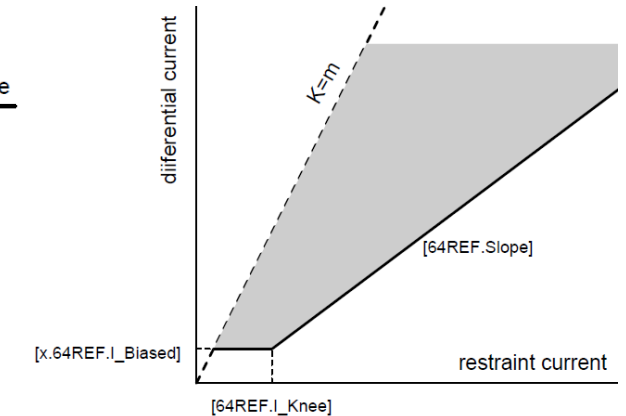
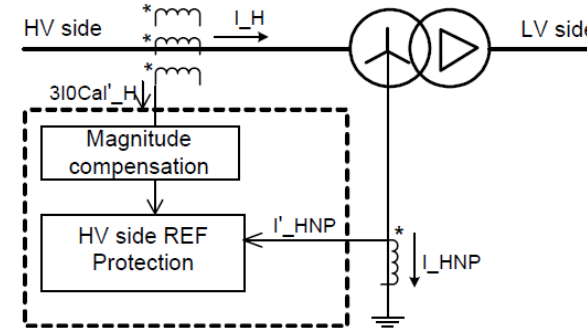
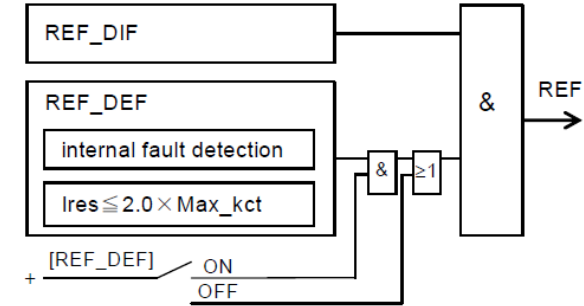
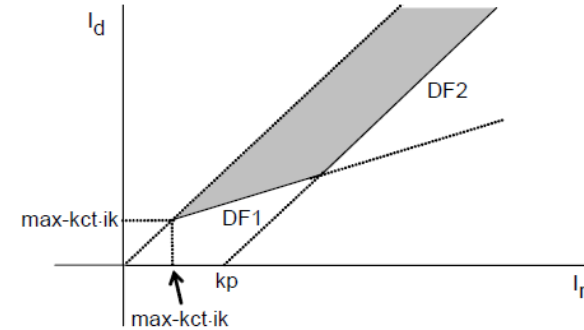


Low Impedance REF Element

Through Fault Stability of REF element is prone to CT saturation.
Different manufacturers require different **blocking** or **restraining** mechanism on such component.

Relay	Differential and Restraint
GRT100	$I_D = K_1 I_{01} + I_N$ $I_R = \max(K_1 I_A , K_1 I_B , K_1 I_C , K_0 I_N)$
PCS978	$I_{0D} = K_1 I_{01} - K_N I_N$ $I_{0R} = \max \{ I_{01}, I_N \}$ * CT Saturation Blocking with 2 nd (for transient) and 3 rd harmonics (for steady state) and CT transient blocking $I_0 < \beta_0 \times I_1$ enabled.
P643#	$K = \text{N-CT Ratio} / \text{ph-CT Ratio}$ $I_D = (I_A + I_B + I_C) + K I_N = 3I_0 + K I_N$ $I_R = 0.5 (\max(I_A , I_B , I_C) + K I_N)$ * Transient Bias and Delay Bias enabled.
7UT613	$I_D = 3I_0 $ $I_R = k (3I_0 - I_N - 3I_0 + I_N)$ * Depending on Phase Relationship

additional restraint + delay timer at operation
= “restraint blocking”



For the flat slope range: $0 \leq I_{bias \max} \leq I_{s1}/K_1$

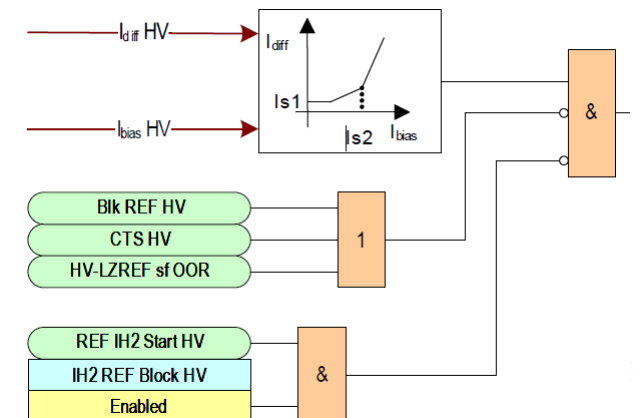
$$I_{diff} \geq I_{s1} + \text{transient bias}$$

For the K1 slope range: $I_{s1}/K_1 \leq I_{bias \max} \leq I_{s2}$

$$I_{diff} \geq K_1 I_{bias \max} + \text{transient bias}$$

For the K2 slope range: $I_{s2} \leq I_{bias \max} \leq I_{s-HS2}/K_2$

$$I_{diff} \geq K_1 I_{s2} + K_2 (I_{bias \max} - I_{s2}) + \text{transient bias}$$



132/11kV Transformer Protection – HSOC Setting

SETTING Criteria:

- At a suitable margin above worst 11kV through fault to ensure stability.
- At a level high enough to ensure stability for worst case inrush current during transformer energization
- At a level sensitive enough to ensure fault detection for the min. fault level condition on 132kV side

I_F = Maximum through fault current

Example

- Maximum Through Fault (11kV side) = 185.2MVA
- Maximum Through Fault Current (11kV side) = $810A = \frac{185.2MVA}{\sqrt{3} (132kV)}$
- 132kV Maximum Fault Level = 6000MVA

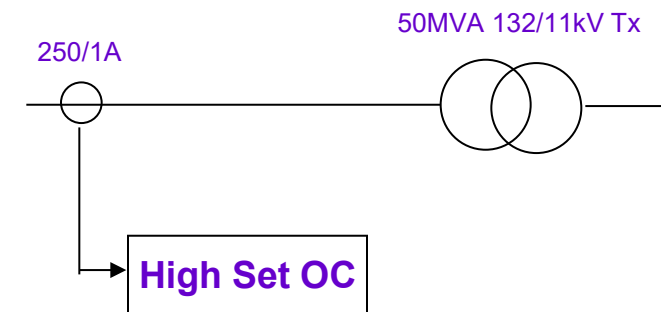
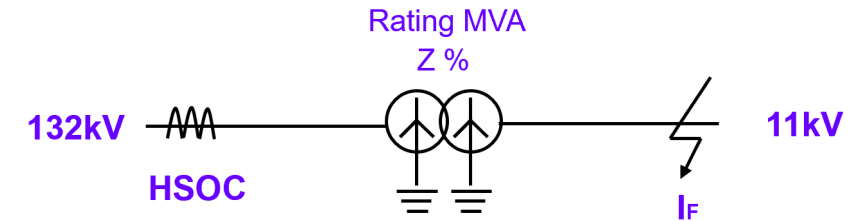
- CT Secondary Current = $810/250 = 3.24A$
- HSOC safety factor is 2.2

- HSOC setting = $3.24 \times 2.2 = 7.12A$

In general, safety factor = 1.3 ~ 1.5 for MHJ, CAG39, KCGG & DCD

= 2 ~ 2.2 for CAG37

(To prevent mal-operation under relay transient overreaching)



$$\text{Safety Factor} \times I_{F,MAX} (11kV) < \text{HSOC Setting} < I_{F,MIN} (132kV)$$

Case 4: Transformer HSOC Tripped on Through Fault

- A transformer was tripped on through fault ($I_F = 766A$) by HSOC (Setting = 788A) on 2005 – 07 – 18.
- It was found that the **maximum through fault at maximum tap** = 680A from system parameter calculation. Hence, the HSOC setting 788A should be acceptable. The existing transformer impedance parameter was in doubt.
- Upon site transformer impedance test conducted on 2005 – 08 – 05, it was found that the transformer impedance values provided from transformer test report was given on base of **tap voltage & current**, instead of **nominal voltage and current** as expected.

Tap Position	Injected HV Current [A]	Injected HV Voltage [V]	Measured Impedance at 30°C [%]	Calculated Impedance with Ref. to Nominal Tap 8 [%]	Factory Test Impedance at 75°C [%]
1	1.334	395.2	28.13	34.36	28.06
8	1.754	395.6	26.16	26.16	26.03
13	2.152	395.7	24.92	21.33	N/A
24	3.507	395.4	22.63	13.08	22.48

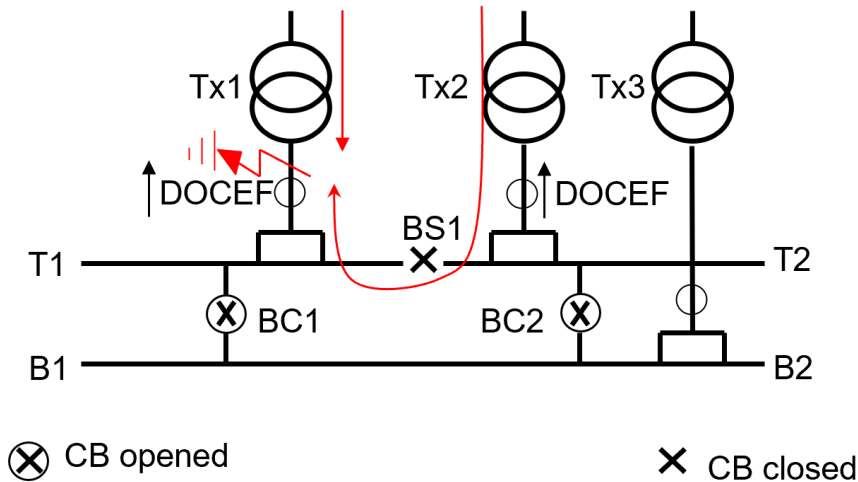
Note –

The calculated Impedance is based on nominal Tap 8 (132kV & 153.1A) as the calculation based for all tappings.

Recall Base Change Formula:

$$Z_2 = Z_1 \left(\frac{V_{1B}}{V_{2B}} \right)^2 = 22.63 \left(\frac{1 - 24\%}{1} \right)^2 = 13.08 \quad \text{Lowest Tap} = -24\%$$

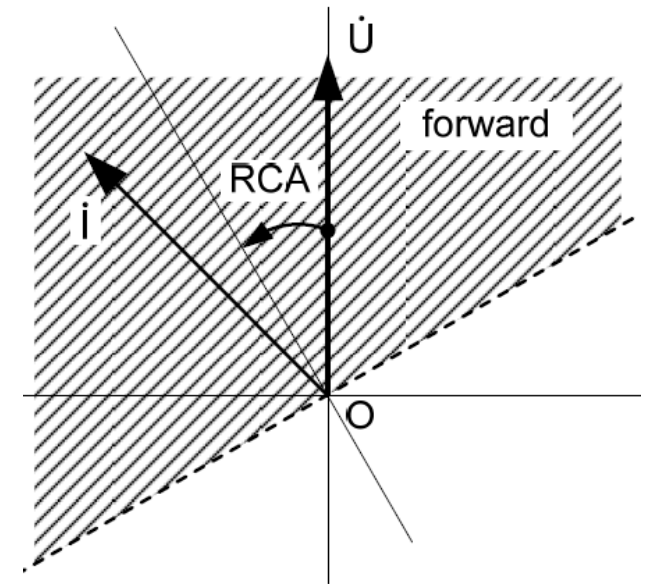
132/11kV Transformer Protection – DOCEF



- **35MVA transformers** are operating in **parallel**
- DOCEF provides **backup** for **uncleared upstream fault** in case **Intertripping** does not work.
- Operate with “forward” current, i.e. current going back to 132kV.
- Direction is given by the comparison between voltage and current phasor
- During VT fuse burnt condition, the voltage phasor becomes unreliable.
- Setting: OC = 0.3A (very sensitive)

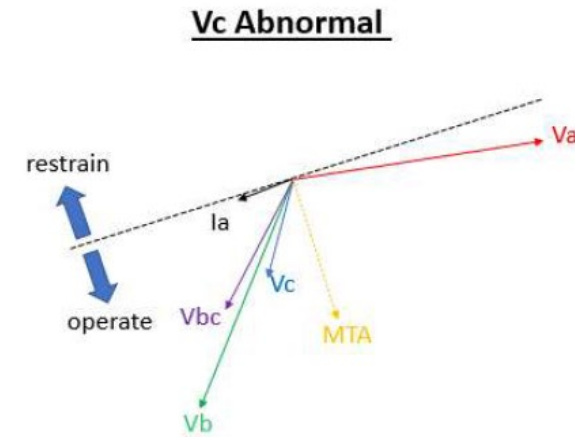
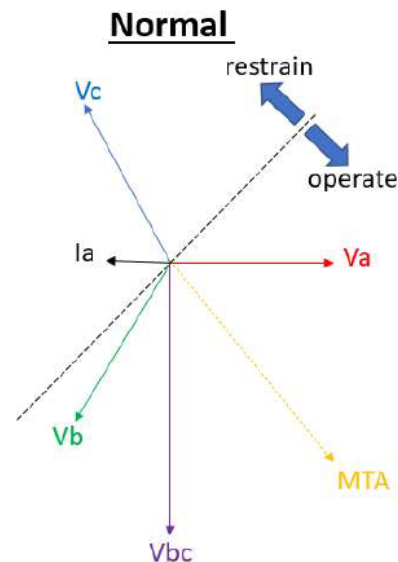
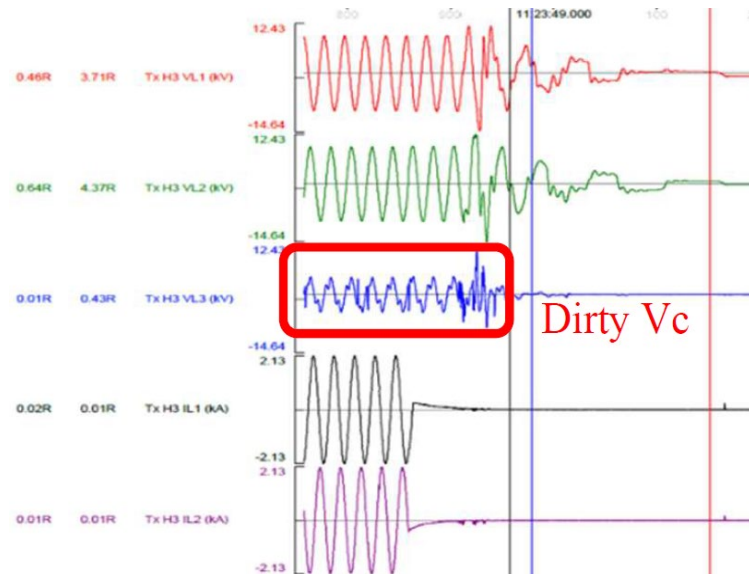
Phase of Protection	Operate Current (I)	Polarizing Voltage (U)
A Phase	I _a	U _{bc}
B Phase	I _b	U _{ca}
C Phase	I _c	U _{ab}

- Directional forward
 $-90^\circ < (\text{angle(I)} - \text{angle(U)} - \text{RCA}) < 90^\circ$
- Directional reverse
 $-90^\circ > (\text{angle(I)} - \text{angle(U)} - \text{RCA}) > 90^\circ$



Case 5: DOCEF Operative with Fuse Burnt

- Prior to the incident, **HV VT fuse** for a transformer circuit was blown out. A planned outage was scheduled for **VT fuse replacement** at 01:00hrs on 28 Jun 2022. At 19:24hrs on 27 Jun 2022, the transformer **tripped on DOCEF protection** and all other ends of the RMU circuit were inter-tripped.
 - From fault recorder waveform retrieved, **dirty voltage signal** with significant deformation in both **magnitude** and **phase angle** could be observed in L3 voltage. Owing to the abnormality in L3 voltage, the forward region of directional element would be shifted accordingly, resulting in the operate current marginally within the forward operation region.
- 1) Request SOD to shut down the tx for VT fuse replacement, or
 - 2) Request SOD to re-configure busbar config so that the Tx with VT problem could run in parallel with other Tx (to cater for DOCEF undesired operation), or
 - 3) Issue a TLR to state “possible unstable of DOCEF protection”, or
 - 4) Defeat the concern DOCEF protection if the tx is running as independent auto (IA) mode, consider it is a backup protection of the tx main protection.



132/11kV Transformer Protection – Buchholz

Gas Production causes oil to be displaced and causes the float to drop and operate the switch for alarm purpose. Another switch is calibrated to operate at a certain oil flow rate (surge) to trip the transformer CB.

Buchholz Gas

- Gas generation may be a result of the following:
 - Decomposition / degrading of solid or liquid insulation inside the transformer due to the overheating or arcing.
 - From the outside towards the pipeline.
 - From the oil itself due to unsatisfactory de-gassing prior filling.
- If sufficient gas is found to displace the upper float, the alarm (BGA) should be activated.
- The small valve at the top left is to bleed the gas off and reset the relay.

Buchholz Surge

- Rapid oil movement in the pipeline towards the conservator is caused by an internal arc short circuit, or hot spot which must be correctly addressed.
- Oil leaks from the transformer are environmentally unacceptable and a fire hazard will lead to transformer failure.

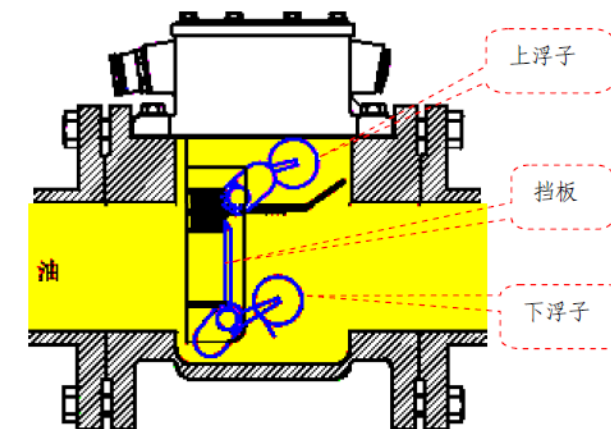


图1-2-1-14 正常情况下，气体继电器状态示意图

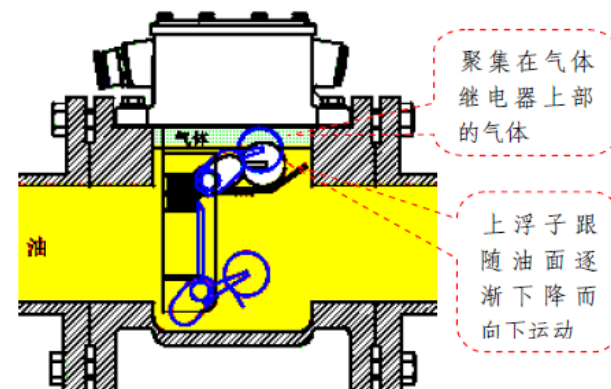
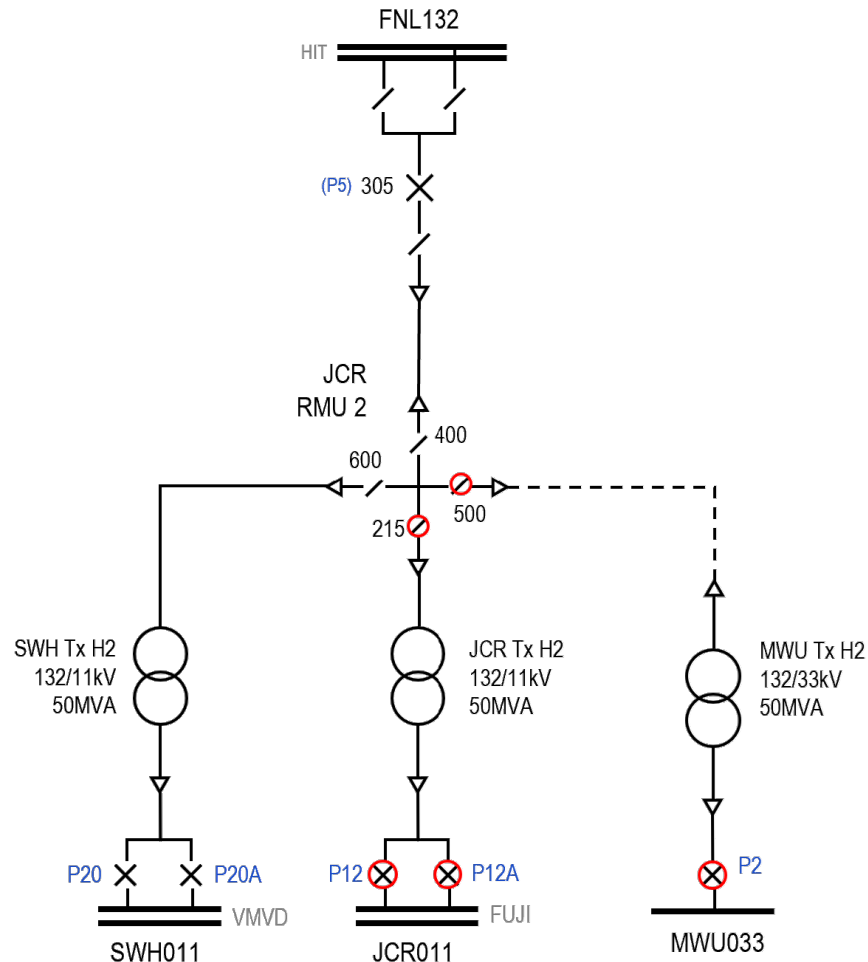


图1-2-1-15 气体继电器内聚集气体示意图

Case 6: Tripping Incident during Tap Changer Maintenance



ITR 1562172-1 Event Summary:

Prior to the incident, JCR Tx H2 and JCR RMU 2 – MWU FT H2 were isolated for cable sealing end replacement and transformer maintenance respectively under planned outage. Yet, transformer SWH011 was fed by FNL132 to JCR RMU 2.

The transbution circuits tripped at 11:31 hrs. Supply to R1 busbar in SWH011 S/S was then restored by numerical auto-switching scheme.

Site investigation revealed that the intertrip send signal was initiated from MWU Tx H2 Tap Changer Buchholz Surge Trip Protection. After all relevant intertrip link were removed, FNL – JCR RMU 2 – SWH FT H2 were put back to service at 13:41.

Protection Operation at MWU Side

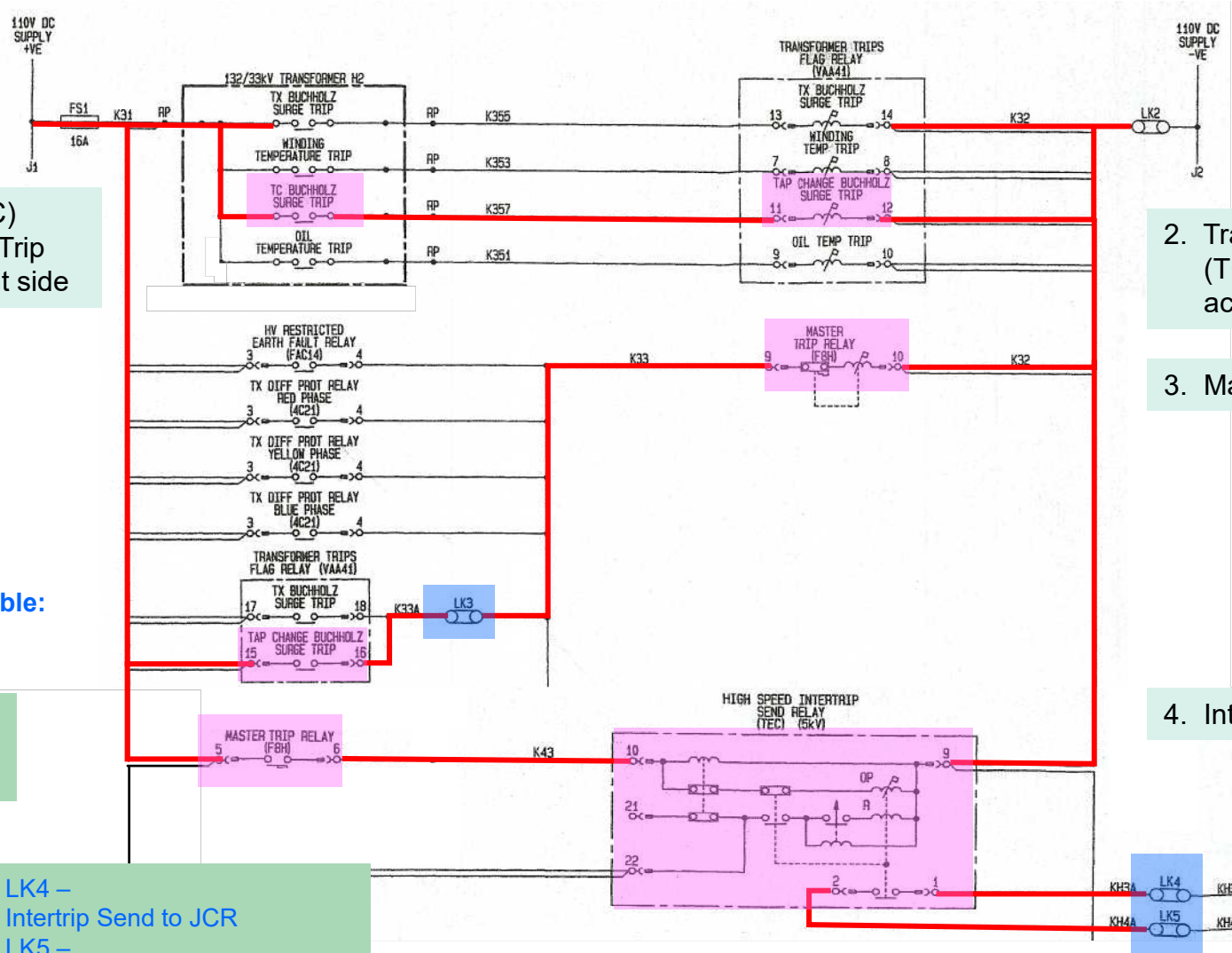
1. Tap Changer (TC) Buchholz Surge Trip Activated at Plant side

Isolation Links Available:



LK3
Buchholz
Surge Trip

LK4 –
Intertrip Send to JCR
LK5 –
Intertrip Receive to JCR

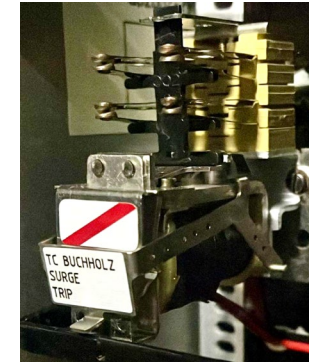


2. Transformer Flag Trip Relay (TC Buchholz Surge Trip) activated

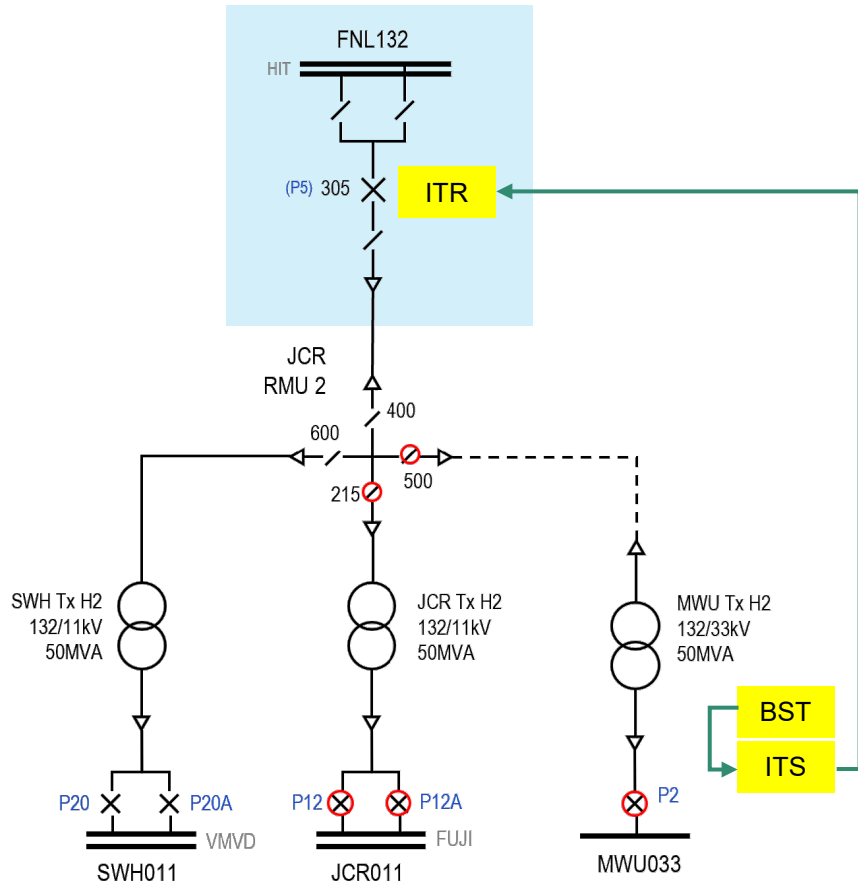
3. Master Trip Relay activated

4. Intertrip Send Relay activated

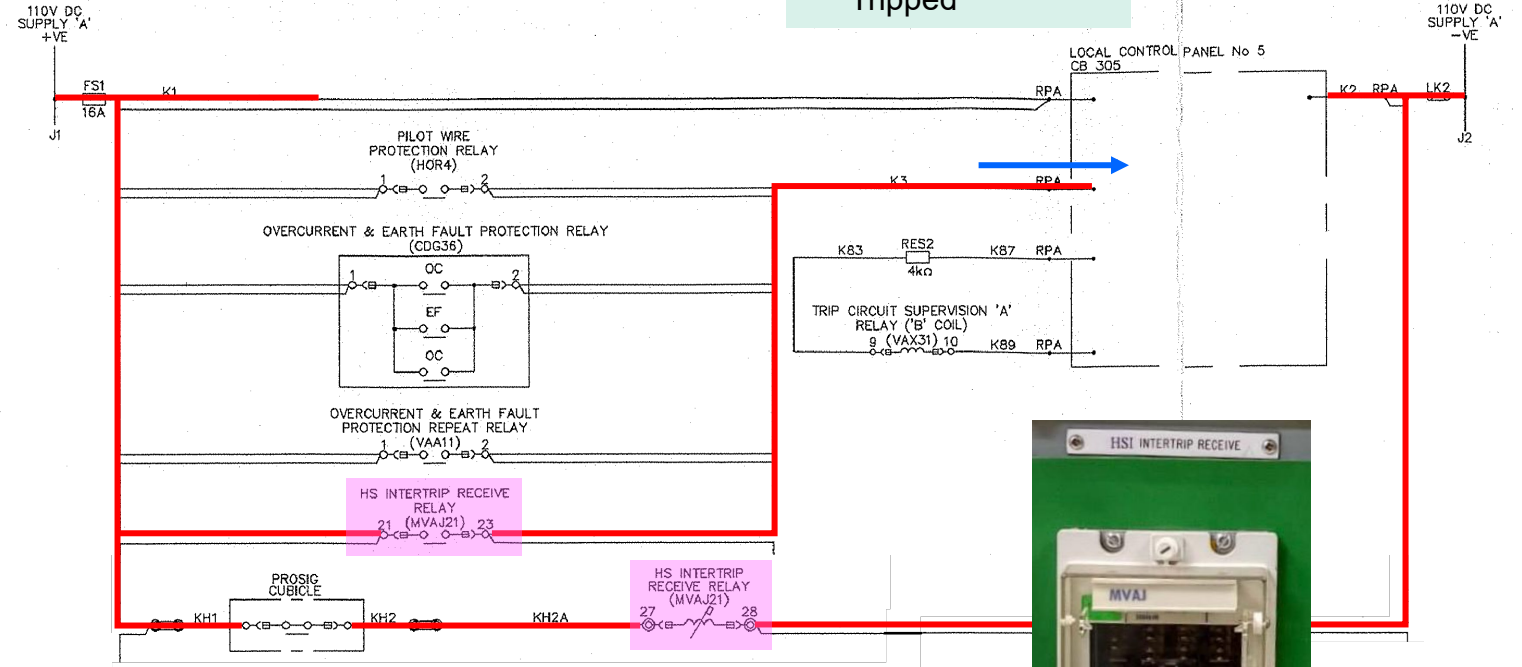
Intertrip to FNL, JCR and SWH



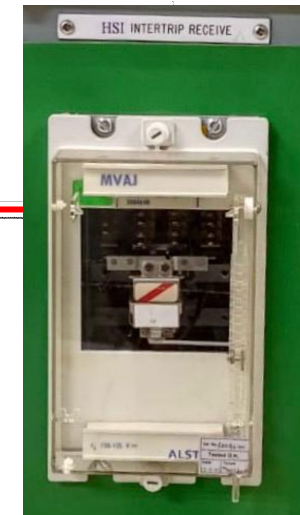
Protection Operation at FNL Side



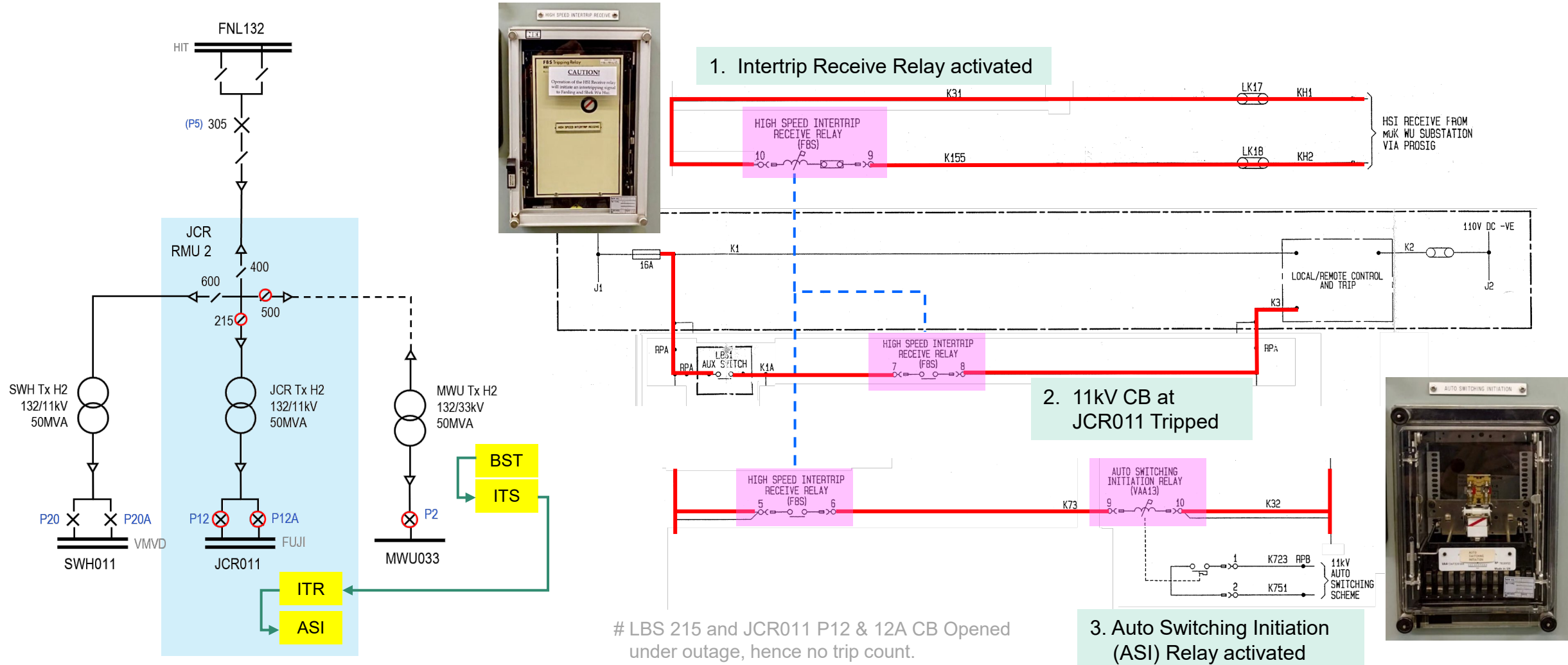
2. CB305 at FNL132 Tripped



1. Intertrip Receive Relay activated



Protection Operation at JCR Side



SWH Tx H2
132/11kV
50MVA

JCR Tx H2
132/11kV
50MVA

MWU Tx H2
132/33kV
50MVA

P20 P20A VMVD
SWH011

P12 P12A FUJI
JCR011

P2
MWU033

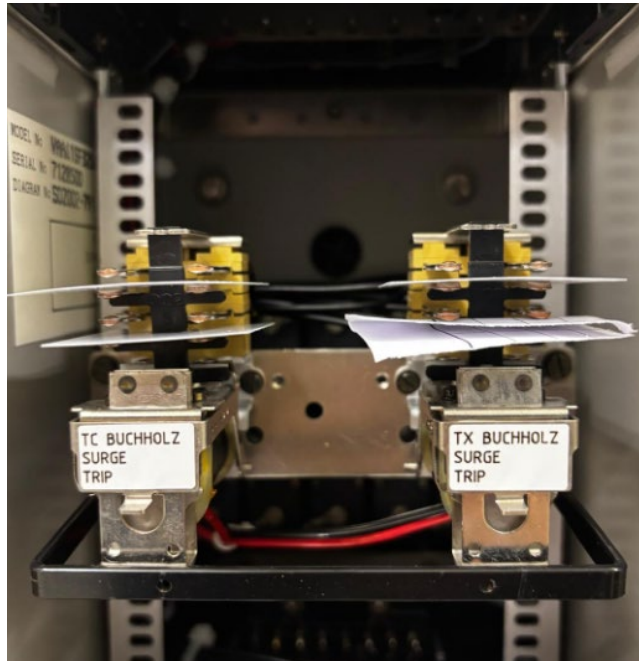
ITR
ASI

BST
ITS



Protection Isolation Performed under Tap Changer Mtc

- To avoid unnecessary inter-tripping activated during tap changer mtc, one should...



Apply Paper Wedge between Contacts
in Transformer Flag Trip Relay
& Alarm Relay (if necessary)



Remove Appropriate Trip
& Intertrip Links

Conclusion

- Percentage Bias and Interposing CT
 - **Case 1: Incorrect Setting Leading to Tripping**
→ Understand the meaning behind symbols and settings, Avoid incorrect decision under urgent restoration.
- Harmonic Bias
 - **Case 2: Transformer Tripped under Energization and De-Energization with Harmonic Blocking**
→ A correct detection algorithm is needed to maintain stable under ultra-saturation and subsidence current.
- Restricted Earth Fault (REF) Element
 - **Case 3: LVREF Unstable Leading to Through Fault Tripping**
→ Understand the quantity and the physics behind protection logic.
- HSOC
 - **Case 4: Transformer HSOC Tripped on Through Fault**
→ Understand the mathematics behind the calculations.
- Directional OCEF as Backup
 - **Case 5: DOCEF Operative with Fuse Burnt**
→ Fully utilize the relay logic to maintain stability
- Buchholz
 - **Case 6: Tripping Incident during Tap Changer Maintenance**
→ Understand the interaction of works between different parties.
- Conclusion



THE END

ANY QUESTIONS?

Power System Analysis and Protection Studies

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